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Analog Integrated Circuit Laboratory

Final Project Report

Section: G2 Group: 03

Design of a CMOS Integer-16 Charge-Pump PLL

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1 Abstract

This report presents the transistor-level design, layout and verification of a CMOS integer-16 charge-pump phase-locked loop that synthesises a 1.20 GHz clock from a 75 MHz reference. The loop is a type-II, third-order architecture built in the Cadence GPDK045 45 nm process at a 1.1 V supply, and every block — tri-state phase-frequency detector, charge pump, passive loop filter, current-starved ring oscillator and divide-by-16 chain — is implemented with PDK devices only. A two-inverter reset-delay cell with 300 nm channels, together with output buffers, removes the PFD dead zone and gives full-swing 225 ps pulses with a linear charge-versus-phase characteristic of 0.0525 fC/ps. Because channel-length modulation limits a simple-mirror charge pump to an 8.1 μA UP/DN mismatch spread and the 1.1 V supply leaves no headroom for cascoding, a replica-bias charge pump with a rail-to-rail two-OTA error amplifier and an on-chip beta-multiplier reference was used instead, reducing the mismatch to $+0.77 / -1.27 \mu\text{A}$ ($\leq 2.5\%$) over the 0.2–0.9 V control range. The seven-stage ring oscillator uses a source-degenerated low-threshold input device to spread 900.7 MHz to 1.50 GHz over that same window, and long-channel bias devices followed by impedance scaling improve its phase noise from -76.8 to -91.9 dBc/Hz at 1 MHz offset. The loop filter was computed for a 3.75 MHz bandwidth with 60° of phase margin and then rescaled for the measured VCO gain of about 1.08 GHz/V. The complete PLL locks in 612.5 ns at 1.200 GHz with 1.214 ps RMS period jitter, a -69.4 dBc reference spur and 2.315 mW of power, and the layout is DRC and LVS clean in an area of 0.6251 mm².

2 Introduction

Modern digital and radio-frequency systems need gigahertz clocks, but quartz crystals, the only inexpensive references with the required stability, resonate at a few tens of megahertz at most. A phase-locked loop (PLL) bridges this gap by producing a high-frequency output locked to a low-frequency reference, and the integer-N charge-pump PLL is the standard form when the output is an integer multiple of the reference [1], [2].

Operating principle. A phase-frequency detector (PFD) compares the reference with a divided copy of the output and produces UP and DN pulses whose width difference is proportional to the phase error. A charge pump turns these pulses into charge, a passive loop filter integrates it into a control voltage, and the voltage-controlled oscillator (VCO) converts that voltage back into frequency. A divide-by- N counter closes the loop, so in lock $f_{\text{out}} = Nf_{\text{ref}}$. The loop filter sets how fast the loop locks and how much reference and VCO noise reaches the output.

The problem addressed here. The project required a 1.20 GHz synthesiser that multiplies a 75 MHz reference by $N = 16$, designed at transistor level in the Cadence GPDK045 45 nm process at 1.1 V. The architecture was fixed: a dead-zone-free tri-state PFD, a switched charge pump, a passive second-order loop filter, a seven-stage current-starved ring VCO and four divide-by-two flip-flops. The loop had to reach a 3.75 MHz bandwidth with 60° phase margin, lock within 8 μs , meet the jitter, phase-noise, spur and power limits, and pass DRC and LVS. The full specification is given in Section 3.1.1.

Why this is a complex engineering problem. The design has the following attributes of a complex engineering problem.

- **Depth of knowledge (P1).** It combines short-channel device physics, analog circuit design,

feedback theory, oscillator noise theory, high-speed digital timing and 45 nm layout rules.

- **Range of conflicting requirements (P2).** VCO gain conflicts with tuning range, dead-zone removal with reference spur, charge-pump matching with the 1.1 V headroom, phase noise with power, and loop bandwidth with die area (Section 3.1.4).
- **Depth of analysis required (P3).** The matching target had to be shown unreachable by sizing alone, and the loop bandwidth and phase margin had to be verified on a separate linear model, because a locked PLL cannot be linearised by an AC analysis.
- **Familiarity of issues (P4).** Problems such as a replica feedback loop nested inside the PLL and charge dumping from switched current sources were met for the first time in this project.
- **Extent of applicable codes (P5).** Every device had to respect the GPDK045 reference and DRC manuals [3], the validity range of the PDK transistor models [4], the 1.1 V supply and the course layout guideline [5].
- **Extent of stakeholder involvement (P6).** The design had to satisfy the course specification, the PDK rules and the eventual user of the clock, who cares about jitter and spurs rather than block-level numbers.
- **Interdependence (P7).** The loop gain contains $I_{CP}K_{VCO}/N$, so the filter could only be computed after the VCO was measured, and changes to the PFD or charge pump altered the ripple, spur and stability of the whole loop.

Alternative solutions. The specification fixed the architecture, but each choice has alternatives.

- **System architecture.** A delay-locked loop cannot multiply frequency, an all-digital PLL needs a time-to-digital converter and a digitally controlled oscillator, and a fractional-N loop is unnecessary because $N = 16$ is an integer. The integer-N charge-pump PLL is the standard choice for this task [1].
- **Phase detection.** An XOR detector needs a 50 % duty cycle, and neither the XOR nor the JK flip-flop detector can detect frequency, whereas the tri-state PFD has a $\pm 2\pi$ linear range, zero static phase error in a type-II loop and intrinsic frequency detection [1], [2].
- **Charge pump.** A simple mirror gave an 8.1 μA mismatch spread, about 1.6 times the allowed $\pm 2.5 \mu\text{A}$ window, and a cascode has no headroom at 1.1 V, so a replica-biased charge pump, in which an error amplifier forces both branches to the same operating point [6], was adopted.
- **Oscillator.** An LC VCO has lower phase noise but needs an inductor and tunes only a few per cent, and a differential (Maneatis) ring [7] needs its own bias loop; the specified current-starved ring is compact and tunes over more than an octave.
- **Frequency divider.** True single-phase clock (TSPC) flip-flops [8] need one clock phase and about nine transistors, so they load the VCO less than static CMOS or current-mode logic.
- **Loop filter.** An active filter would allow a smaller capacitor but adds noise in the control path; the specified passive $R-C_1 \parallel C_2$ filter keeps the control node quiet at the cost of area.

Section 3 formulates the problem, derives the design equations and presents the layout; Section 4 presents the verification results and the assessment of the design; and Sections 5–8 cover teamwork, communication, project management and future work.

3 Design

3.1 Problem Formulation (PO(b))

This section states what had to be designed and what was deliberately left out, reviews the published work the design builds on, turns the system specification into a set of block-level targets, and analyses those targets to find where they conflict — before any transistor was sized.

3.1.1 Identification of Scope

The object of the design. A type-II, third-order charge-pump PLL that produces a 1.20 GHz output locked to a 75 MHz reference, built from five blocks (PFD, charge pump, loop filter, VCO and divider) in one feedback loop. The targets allocated to this group and the requirements common to all groups are listed in Table 1.

Parameter	Target	Set by
Reference frequency, f_{ref}	75 MHz	Group table
Division ratio, N	16 ($4 \times \div 2$)	Group table
Output frequency, f_{out}	1.20 GHz	Group table
VCO topology	7-stage single-ended current-starved ring	Group table
VCO tuning range (minimum)	900 MHz to 1.50 GHz	Group table
VCO gain, K_{VCO}	750 MHz/V	Group table
Charge pump current, I_{CP}	50 μA	Group table
Loop bandwidth	3.75 MHz	Group table
Phase margin	60°	Group table
Lock time	$\leq 8 \mu\text{s}$	Group table
Technology	GPDK045 PDK, Cadence Virtuoso with Spectre	All groups
Supply voltage, VDD	1.1 V	All groups
Design level	Every block at transistor level	All groups
Reference clock	An ideal vpulse source may be used	All groups
PFD	Tri-state, free of a dead zone	All groups
Charge-pump matching	UP and DN within 5 % for $V_{\text{c}} = 0.2\text{--}0.9 \text{ V}$	All groups
VCO tuning-range rule	Must cover $f_{\text{out}} \pm 25 \%$ over the control-voltage range	All groups
RMS period jitter	$\leq 8 \text{ ps}$	All groups
VCO phase noise at 1 MHz offset	$\leq -90 \text{ dBc/Hz}$	All groups
Reference spur	$\leq -40 \text{ dBc}$	All groups
Total power consumption	$\leq 5 \text{ mW}$	All groups

Table 1: Design specifications for the PLL: the targets allocated to this group and the requirements common to all groups.

What the project includes.

- Transistor-level design of all five blocks and of an on-chip current reference, using GPDK045 devices only.
- Calculation of the loop filter from the block gains, and verification of the loop bandwidth and phase margin.
- Pre-layout verification in Spectre at the typical corner and $V_{\text{DD}} = 1.1 \text{ V}$.
- Full custom layout of every block and of the top level, clean in DRC and LVS.

What the project excludes.

- The reference oscillator (an ideal vpulse source is permitted), supply regulation, ESD protection, pads and packaging.
- Post-layout, corner, supply and temperature simulation; every result is pre-layout, typical-corner and nominal-supply.
- Fabrication and measurement, and programmable or fractional division, which $N = 16 = 2^4$ does not need.

Constraints that bound the design space. The 1.1 V supply rules out cascoding in the charge pump at low control voltages. The VCO must cover $f_{out} \pm 25\%$, while the charge pump need only match over $V_c = 0.2\text{--}0.9$ V, so the whole 600 MHz span must fit inside that 0.7 V window. The PDK models are valid for widths up to 10 μm per finger and lengths down to 40 nm [4], and the loop filter must use resnspoly and mimcap devices, whose values come in fixed steps.

3.1.2 Literature Review

The design builds on established results. The review below is organised by the question each source answers.

Loop theory and the linear model. Razavi [1] gives the phase-domain model used in Section 3.2: the PFD and charge pump form a gain of $I_{CP}/2\pi$, the VCO an integrator $2\pi K_{VCO}/s$ and the divider a gain $1/N$. A type-II loop needs a filter zero for stability and a third pole to suppress ripple, and the model holds only if the loop bandwidth is at least about ten times below the reference frequency [2].

Loop-filter design procedure. Fischette [9] and Palermo [10] give the procedure used here: place the third pole at a multiple of the crossover, set the zero from the remaining phase margin, and find the total capacitance from the unity-gain condition. Jha's report [11] provided a worked example that was used as a cross-check.

The dead zone in a tri-state PFD. For small phase errors the reset path can clear the flip-flops before the charge-pump switches turn on, so the loop delivers almost no charge. The remedy is a deliberately longer reset path, at the cost of charge injected by any UP/DN mismatch during the overlap [1], [9].

Charge-pump matching and reference spurs. Channel-length modulation drives the UP and DN currents apart as the control voltage moves. Of the published remedies, only replica biasing, in which an error amplifier forces a replica branch to track the control voltage [6], works at 1.1 V. The residual mismatch and the switch charge injection set the reference spur [9].

On-chip current reference. The beta-multiplier (constant- g_m) reference sets its current from a resistor and a device ratio, largely independent of the supply, but needs a start-up circuit [1], [12]. It is used in Section 3.2.2.

Ring-oscillator phase noise. Hajimiri and Lee [13] show that flicker noise in the bias devices is upconverted around the carrier, so enlarging their gate area lowers the flicker corner at no speed cost, and that the thermal phase noise of a ring falls in proportion to its power [14]. Scaling all widths by k and all bias resistances by $1/k$ therefore gains about $10 \log_{10} k$ dB at k times the power.

Current-starved ring VCO. Driving the starving devices directly from the control voltage gives a steep, non-linear tuning curve; source degeneration and a fixed offset current lower and linearise it [12]. Both are used in Section 3.2.3.

Divider logic. TSPC logic, introduced by Yuan and Svensson [8], gives an edge-triggered flip-flop

with one clock phase and about nine transistors, but its dynamic nodes impose a minimum operating frequency.

Process and layout rules. The GPD045 reference and DRC manuals [3] define the available devices, the PDK model report [4] gives their valid geometry range, and the course layout guideline [5] fixes the drawing and routing conventions.

What the literature does not settle. Published procedures treat K_{VCO} as a free choice. Here the tuning range and K_{VCO} are specified independently and cannot both be met (Section 3.1.4); rescaling the filter for the measured gain is the one step not taken from a published procedure.

3.1.3 Formulation of Problem

Statement of the problem. Find transistor-level implementations of the five blocks, and values of R , C_1 and C_2 , such that the loop locks to $f_{out} = Nf_{ref} = 1.20$ GHz from any starting condition with a 3.75 MHz crossover and 60° phase margin, while meeting the limits of Table 1 and passing DRC and LVS.

The loop equation. The open-loop gain is the product of $I_{CP}/2\pi$, the filter impedance $Z(s)$, $2\pi K_{VCO}/s$ and $1/N$. A series R – C_1 branch in parallel with C_2 adds a pole at the origin, a zero and a high-frequency pole, which gives the required type-II, third-order loop. With I_{CP} and N fixed and K_{VCO} measured, R , C_1 and C_2 follow from three conditions: unity gain at 3.75 MHz, 60° phase margin and the position of the third pole (Section 3.2.4).

Decomposition into block-level targets. Table 2 translates the system specification into a target for each block and the quantity that block delivers to the loop equation.

Block	Function in the loop	Specified requirement	Delivered to the loop
Phase-frequency detector	Turns the phase error between reference and feedback into an UP/DN pulse-width difference	Tri-state; two D flip-flops with asynchronous reset; free of a dead zone	Gain $I_{CP}/2\pi = 7.96$ $\mu\text{A}/\text{rad}$, linear through zero phase error
Charge pump	Converts the UP/DN pulses into a net charge on the filter	$I_{CP} = 50$ μA ; $ I_{UP} - I_{DN} \leq 2.5$ μA for $V_c = 0.2$ – 0.9 V; on-chip current reference	The 50 μA that sets the loop gain; the matching that sets the spur
Loop filter	Integrates the charge into the control voltage and sets the loop dynamics	Passive: series R – C_1 branch in parallel with C_2	$Z(s)$: a zero, a pole at the origin and a high-frequency pole, giving $f_c = 3.75$ MHz at PM = 60°
Voltage-controlled oscillator	Converts the control voltage back into frequency and integrates it into phase	7-stage single-ended current-starved ring; 0.90–1.50 GHz over $V_c = 0.2$ – 0.9 V; $K_{VCO} = 750$ MHz/V; phase noise ≤ -90 dBc/Hz at 1 MHz	Gain $2\pi K_{VCO} / s$
Frequency divider	Closes the loop by dividing the output back to the reference rate	Four cascaded divide-by-two flip-flops ($2^4 = 16$); must toggle at 1.50 GHz	Gain $1/N = 0.0625$
Complete loop	Produces the locked output clock	Lock ≤ 8 μs ; RMS period jitter ≤ 8 ps; reference spur ≤ -40 dBc; power ≤ 5 mW; DRC- and LVS-clean layout	$f_{out} = N f_{ref} = 1.20$ GHz

Table 2: Decomposition of the system specification into a target for each block and the quantity that block delivers to the loop equation.

Conditions that must hold in addition to the gains. Four further conditions follow from the specification and were used as sizing rules.

1. **Dead-zone condition.** The minimum PFD pulse must outlast the switch turn-on time, $t_{reset} > t_{on}$.
2. **Charge-pump output resistance.** Keeping the mismatch below 2.5 μA over $\pm 0.35\text{ V}$ requires $r_o \geq 2\Delta V/\Delta I \approx 280\text{ k}\Omega$ for each source.
3. **Ring stage delay.** Seven stages need a stage delay of 47.6 ps at 1.50 GHz and 79.4 ps at 0.90 GHz.
4. **Divider timing.** The first stage needs clock-to-Q delay plus setup time below 667 ps (1/1.50 GHz); later stages run slower.

3.1.4 Analysis

Before any device was sized, the targets of Table 2 were examined to find which are easy, which are binding and which conflict.

The loop-level numbers are self-consistent. Since $f_{ref}/f_c = 20$, the s-domain model holds. The unity-gain condition gives about 13–18 pF of filter capacitance, and slew-limited acquisition plus linear settling give a minimum lock time of about 0.5 μs , far inside the 8 μs limit.

The VCO gain and the tuning range cannot both be met. Covering 600 MHz inside the 0.7 V matched window needs at least 857 MHz/V on average, already above the specified 750 MHz/V, and the curved f–V characteristic makes the local slope higher still.

The tuning range was kept. It is a hard requirement, whereas K_{VCO} enters the loop only through $I_{CP}K_{VCO}/N$. Scaling both capacitors up and the resistor down by the gain error restores the crossover without moving the zero, pole or phase margin, so K_{VCO} was measured on the finished VCO (Section 3.2.3) and the filter was computed last (Section 3.2.4).

Charge-pump matching is an architectural problem. Matching to 2.5 μA needs about 280 k Ω per source, while a 50 μA device in 45 nm gives a few tens of kilohms; even a 4 μm mirror left about 8 μA of mismatch. Cascoding has no headroom at 1.1 V with V_c down to 0.2 V, so the currents had to be equalised actively by a replica-bias loop (Section 3.2.2).

That replica loop is a second feedback loop inside the PLL whose gain changes with V_c , so it must be stable over the whole 0.2–0.9 V range (Section 4.3).

Removing the dead zone costs reference spur. During the minimum UP/DN overlap any mismatch injects $\Delta Q = \Delta I \times t_{reset}$ every cycle, which ripples the control voltage and creates sidebands at $f_{out} \pm f_{ref}$. The reset delay was therefore made only as long as needed (Section 3.2.1), and the third filter pole adds attenuation at f_{ref} .

Phase noise has to be bought with power. Once the flicker corner is pushed below 1 MHz by enlarging the bias devices, the remaining thermal noise improves only by $10 \log_{10} k$ dB for k times the power. Closing the 13 dB gap of the first oscillator therefore needed about ten times more VCO power, which the 5 mW budget allows.

Area is set by the loop bandwidth. The 13–18 pF filter capacitance is fixed by I_{CP} , K_{VCO} , N and the crossover frequency, so the mimcap array was accepted in advance as the dominant area, about 89 % of the combined block area.

Speed is not the difficulty. A minimum-length inverter is far faster than the 47.6 ps stage delay

needed, so the ring has to be deliberately starved, and the TSPC divider has ample margin against its 667 ps input period.

Implications for verification. A locked PLL has no DC operating point, so the bandwidth and phase margin are measured on a linear model built from the measured block gains, lock, ripple, spur, jitter and power are measured in closed-loop transient, and each block is checked against its row of Table 2 (Section 4.3.1).

Summary of the conflicts. Table 3 lists the conflicting requirements, why they conflict and the resolution adopted for each.

Conflicting requirements	Why they conflict	Resolution adopted
$K_{VCO} = 750 \text{ MHz/V}$ vs tuning range 0.90–1.50 GHz over $V_c = 0.2\text{--}0.9 \text{ V}$	600 MHz across a 0.7 V window needs at least 857 MHz/V on average, and the $f\text{--}V$ curve of a starved ring is not straight	Tuning range kept; K_{VCO} measured on the finished VCO and the loop filter rescaled so that f_c and PM are preserved (Sections 3.2.3, 3.2.4)
Charge-pump matching $\leq 5\%$ vs $VDD = 1.1 \text{ V}$ with V_c down to 0.2 V	Matching needs $r_o \geq 280 \text{ k}\Omega$, which sizing cannot reach, and cascoding has no headroom at 1.1 V	Replica-bias charge pump with a rail-to-rail two-OTA error amplifier and a beta-multiplier reference (Section 3.2.2)
Stability of the replica loop vs the PLL loop	The matching circuit adds a second feedback loop inside the PLL whose gain varies with V_c	Compensated for adequate phase margin across the whole 0.2–0.9 V range and verified by stb analysis (Sections 3.2.2, 4.3)
No PFD dead zone vs reference spur $\leq -40 \text{ dBc}$	The reset delay that removes the dead zone also lets any UP/DN mismatch inject charge on every reference cycle	Shortest reset delay that still gives full-swing pulses, built from two long-channel inverters; the third filter pole adds attenuation at f_{ref} (Sections 3.2.1, 3.2.4)
Phase noise $\leq -90 \text{ dBc/Hz}$ at 1 MHz vs total power $\leq 5 \text{ mW}$	Thermal phase noise of a ring oscillator falls only in proportion to power	Flicker corner lowered first with device area at no power cost, then impedance scaling spends the available budget on the oscillator (Section 3.2.3)
Loop bandwidth 3.75 MHz with $I_{CP} = 50 \text{ }\mu\text{A}$ vs die area	The unity-gain condition fixes the filter capacitance at 13–18 pF, which must be built from MIM capacitors	Accepted as a consequence of the specification; the capacitor array dominates the layout (Sections 3.2.4, 3.5)

Table 3: Conflicting requirements identified during problem formulation and the resolution adopted for each.

The blocks were therefore designed and verified one at a time, the VCO gain was measured rather than assumed, the loop filter was computed last, and the loop was then closed and verified in transient. Section 3.2 follows that order.

3.2 Design Method (PO(a))

The PLL was designed block by block from the specifications and then closed at system level. Each block was sized from its governing device equations so that it delivers the gain the loop calculation assumes. The PFD and charge pump provide a gain of $I_{CP}/2\pi$ with $I_{CP} = 50 \text{ }\mu\text{A}$, the VCO provides K_{VCO} , and the divider provides $1/N$, where

$$N = \frac{f_{out}}{f_{ref}} = \frac{1.2 \text{ GHz}}{75 \text{ MHz}} = 16 = 2^4 \quad (3.1)$$

The loop filter was then calculated from these gains to set the loop bandwidth and phase margin. It was calculated first with the specified K_{VCO} , and again with the value measured on the transistor-level VCO. This section gives the reasoning and design equations for each block in that order. The

simulations that verify them are presented in Section 4.3.

3.2.1 Phase-Frequency Detector

Architecture. The PFD is the tri-state type required by the specification [1]. It consists of two D flip-flops with asynchronous reset, both with D tied to VDD. The 75 MHz reference clocks the upper flip-flop, whose output Qa becomes UP. The divider output clocks the lower flip-flop, whose output Qb becomes DN. An AND gate of Qa and Qb drives the reset of both flip-flops. Two further parts were added during the design: a delay cell in the reset path and a two-inverter buffer on each output (Figure 1).

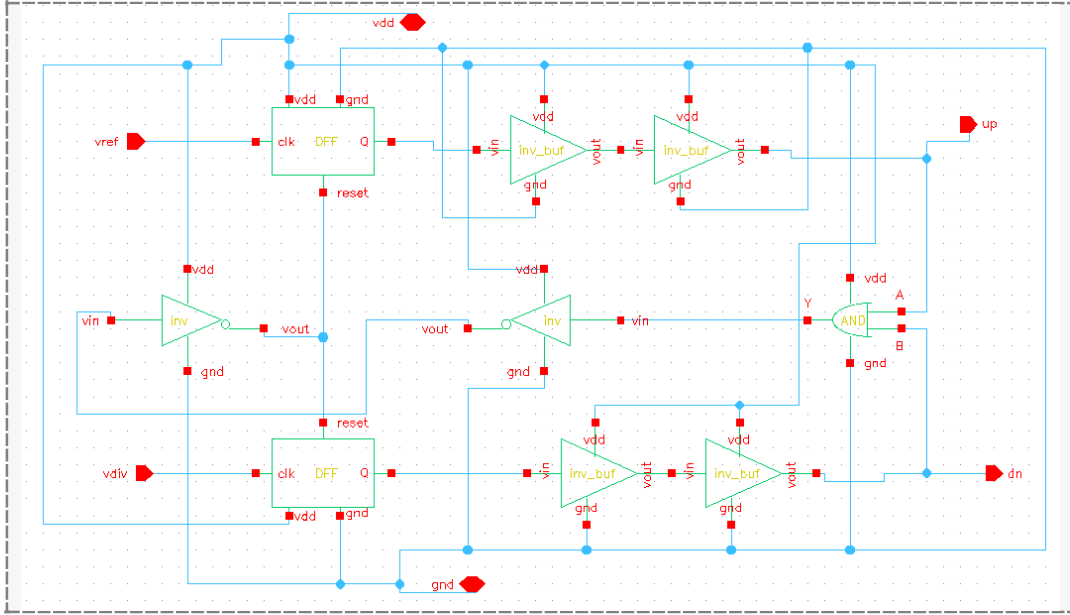


Figure 1: Transistor-level schematic of the tri-state PFD with reset delay cell and output buffers.

Operation and gain. A reference edge sets UP and the charge pump starts sourcing current. When the divider edge arrives, DN is also set. The AND output then goes high and resets both flip-flops, so both outputs return to zero. The circuit therefore has only three states: UP, DN, or neither. If the reference leads by Δt , the UP pulse is wider than the DN pulse by exactly Δt . The net charge delivered per reference period is

$$Q_{net} = I_{CP} \Delta t = I_{CP} \frac{\Delta \phi}{2\pi} T_{ref} \Rightarrow \bar{I}_{CP} = \frac{Q_{net}}{T_{ref}} = \frac{I_{CP}}{2\pi} \Delta \phi \quad (3.2)$$

This is the $I_{CP}/2\pi$ gain used in the loop model. For $I_{CP} = 50 \mu\text{A}$, the ideal slope is 0.05 fC per ps of phase error. If the VCO frequency is wrong, the edges of one input keep arriving first, so one output is active far more often than the other. This also makes the circuit a frequency detector, so the loop can acquire lock from any starting frequency.

Dead zone. In the first version, the AND output drove the reset pins directly. The reset path was then only the AND delay plus the flip-flop reset delay, a few tens of picoseconds in this process. Near lock, the UP and DN pulses were triangular runt pulses about 150 ps wide that did not reach full logic level (DN peaked at about 0.88 V). Such pulses cannot fully turn on the charge-pump switches, which need a finite time $t_{on,CP}$ to turn on and settle at 50 μA . For small phase errors, almost no charge is delivered. The loop is blind to these errors (the dead zone) [1], [9], which causes extra jitter and a nonlinear gain near lock. The specification requires the PFD to be free of a dead zone, and the condition for this is

$$t_{reset} = t_{AND} + t_{delay} + t_{FF,reset} > t_{on,CP} \quad (3.3)$$

Reset delay cell. A two-inverter delay cell was therefore inserted between the AND output and the reset inputs. Even at zero phase error, both UP and DN now stay high for a guaranteed minimum time every cycle, and both switches turn fully on. A phase error only makes one pulse longer than the other, so the net charge stays proportional to Δt all the way down to zero. An even number of inverters was used so that the reset polarity is unchanged. The delay was not made longer than needed, because during t_{reset} both currents flow together, and any UP/DN mismatch ΔI injects an error charge every cycle:

$$Q_{err} = \Delta I \cdot t_{reset} \approx 1.27 \mu\text{A} \times 225 \text{ ps} \approx 0.29 \text{ fC} \quad (3.4)$$

This charge produces ripple on V_c and hence a reference spur. In the complete PLL, the ripple on V_c in lock was 10.13 mV peak-to-peak. The final pulse at lock is about 225 ps wide, only $225 \text{ ps} \times 75 \text{ MHz} \approx 1.7\%$ of the reference period, so the PFD keeps almost its full $\pm 2\pi$ linear range.

Longer channel length in the delay cell ($L = 300 \text{ nm}$). A minimum-length (45 nm) inverter adds only a few picoseconds of delay. Many such inverters would be needed. Instead, the channel length of the two delay inverters was increased. The inverter delay is approximately

$$t_d \approx \frac{C_{load} V_{DD}}{2I_D}, \quad I_D \propto \frac{W}{L}, \quad C_{load} \propto WL C_{ox} \Rightarrow t_d \propto L^2 \quad (3.5)$$

Increasing L both weakens the driving inverter and increases the gate capacitance it must charge, so the delay grows roughly as L^2 . Two inverters with $L = 300 \text{ nm}$ give the required delay in very little area. A longer channel also makes the delay less sensitive to process variation, because a fixed variation ΔL is a smaller fraction of L , and it reduces short-channel effects and leakage.

Output buffers. In the first version, Qa and Qb drove the AND gate and also the gates of the charge-pump switches directly. The switches are large devices and present a heavy capacitive load to the weak flip-flop outputs. A two-inverter buffer (cell inv_but_pfd) was added on each output for three reasons:

- *Full-swing, sharp pulses.* The buffer can drive the switch gates rail to rail with fast edges, so the switches spend little time partially on.
- *Reset timing isolated from the load.* The flip-flop now drives only one small inverter, so the reset loop timing, and therefore the minimum pulse width, is set by the PFD alone. It no longer depends on the charge-pump load, and UP and DN behave identically.
- *Protection against kickback.* Charge injection and clock feedthrough from the switches are blocked from reaching the flip-flop nodes.

Two inverters are used so the output polarity is unchanged. With both modifications, UP and DN at lock are full-swing (0–1.1 V) pulses about 225 ps wide. The dead-zone-free transfer characteristic is verified in Section 4.3.3.

3.2.2 Charge Pump

Specification. The charge pump must deliver $I_{CP} = 50 \mu\text{A}$ through switched PMOS (UP) and NMOS (DN) current sources, with the two currents within 5 % (2.5 μA) of each other for $V_c = 0.2\text{--}0.9 \text{ V}$. Every block must be at transistor level, so the ideal 50 μA reference also had to be replaced by an on-chip circuit.

Why a simple current mirror is not enough. The first version was a PMOS source and an NMOS sink with series switches, mirrored from an ideal reference. A current source in saturation follows

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (3.6)$$

As V_c moves, the V_{DS} of the PMOS source and of the NMOS sink change in opposite directions, so the two currents diverge. Measured from the centre of the range (± 0.35 V), keeping the mismatch below $2.5 \mu\text{A}$ requires

$$\Delta V \left(\frac{1}{r_{o,p}} + \frac{1}{r_{o,n}} \right) \leq 2.5 \mu\text{A} \Rightarrow r_o \geq \frac{2 \times 0.35 \text{ V}}{2.5 \mu\text{A}} = 280 \text{ k}\Omega \quad (3.7)$$

for each device. The mirror length was swept up to $L = 4 \mu\text{m}$, but the best mismatch spread was still $8.1 \mu\text{A}$. The output resistance was about six times too low. Cascoding would raise r_o , but with a 1.1 V supply and V_c down to 0.2 V there is no headroom for a stacked device. Sizing alone could not meet the specification, so the architecture was changed.

Replica-bias charge pump. Since the mismatch comes from unequal V_{DS} , the solution is to force the conditions to be equal [6] (Figure 2). A replica branch (replica PMOS source PMR, replica switch and replica NMOS sink) copies the output branch and is always on. An error amplifier compares the replica output node with V_c and drives the common gate v_{gp} of the UP source PM0 and of PMR. The amplifier adjusts v_{gp} until the replica node equals V_c . The replica then sees exactly the same voltages as the real output, so the PMOS current is corrected to equal the NMOS current at whatever voltage the loop filter holds. The residual mismatch is reduced by the amplifier loop gain A :

$$\Delta I \approx \frac{\Delta I_{\text{open}}}{1+A} \quad (3.8)$$

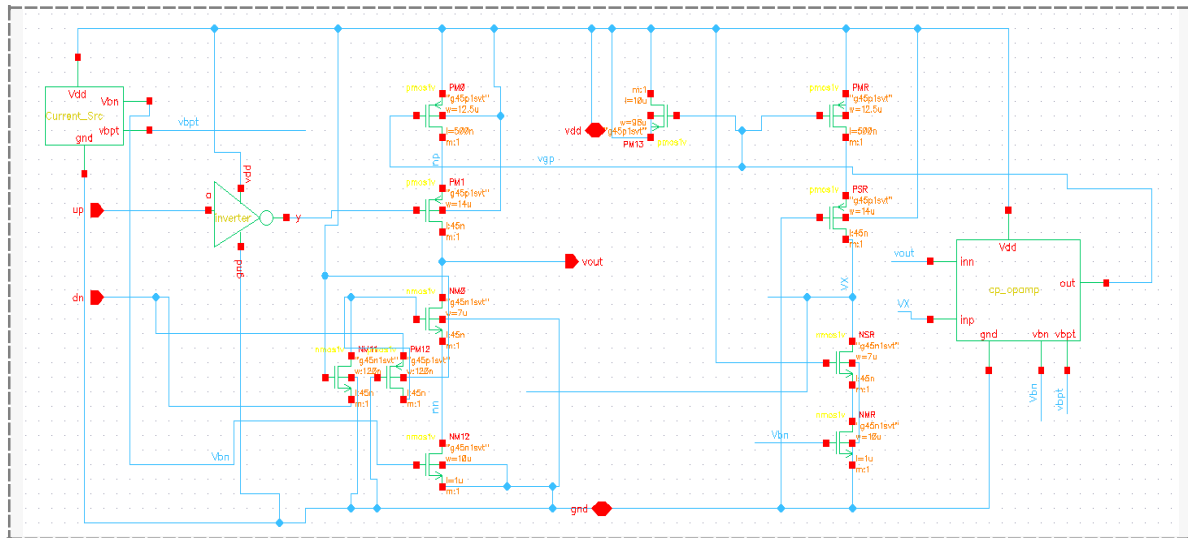


Figure 2: Replica-bias charge pump schematic.

Two-OTA error amplifier. V_c spans 0.2 – 0.9 V, nearly rail to rail. An NMOS-input OTA stops working at low input voltage and a PMOS-input OTA stops working at high input voltage. The amplifier (cell cp_opamp) therefore uses one OTA of each type with their outputs tied, so at least one of them is active over the whole range (Figure 3).

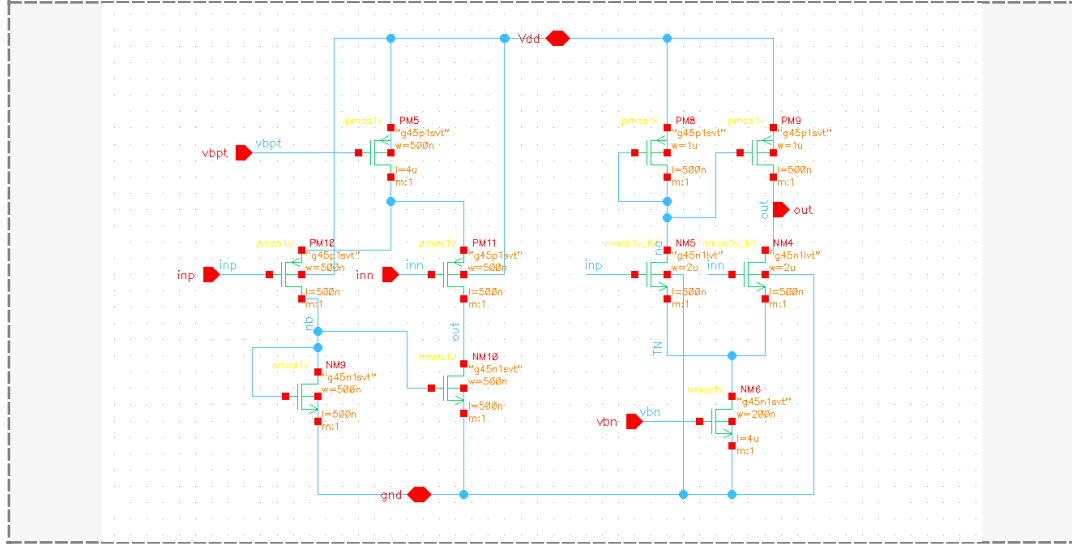


Figure 3: Two-OTA error amplifier used in the replica loop.

Current-source length and charge dumping. With $L = 4 \mu\text{m}$, the large capacitance of the current sources caused a charge dump of about 400 fC each time a switch turned back on, because the source drain node collapses into triode while the switch is off. Once matching was handled by the replica loop, long channels were no longer needed for r_o . The current sources were therefore shortened to $L = 1 \mu\text{m}$, which reduced their area 16 times and largely removed the charge dumping.

On-chip current reference. The ideal $50 \mu\text{A}$ source was replaced by a beta-multiplier (constant- g_m) reference [12], cell Current_Src (Figure 4). It uses a diode-connected NMOS NM4 ($10 \mu\text{m} / 1 \mu\text{m}$), a second NMOS NMB2 that is $K = 4$ times larger with source resistor R_S , and a PMOS mirror PMB3/PMB4 ($25 \mu\text{m} / 1 \mu\text{m}$) that forces both branches to carry the same current. Equating the two branch currents gives

$$I_{REF} = \frac{2}{\mu_n C_{ox} (W/L)_N R_S^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2, \quad \text{for } K = 4: \quad I_{REF} = \frac{1}{2 \mu_n C_{ox} (W/L)_N R_S^2} \quad (3.9)$$

The current depends on R_S and the device size but, to first order, not on VDD, which is why this topology was chosen. R_S is a $2 \mu\text{m}$ wide resnspoly resistor, and its length sets the current. A beta multiplier also has a stable zero-current state, so a start-up circuit (MS1–MS3) was added to make sure the circuit always turns on. A $1 \mu\text{A}$ branch (NM14, PM2) generates the tail bias for the PMOS-input OTA.

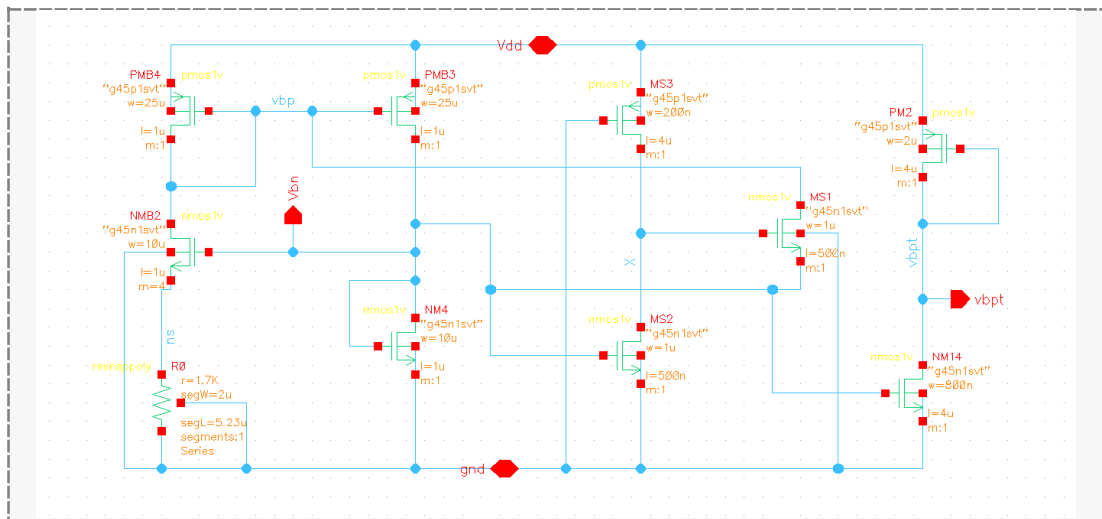


Figure 4: Beta-multiplier current reference with start-up circuit.

Stability of the replica loop. The replica branch and the error amplifier form a feedback loop of their own, which must be stable at every V_c . In mid-range both OTAs are active and their transconductances add, which gives the highest loop gain and the highest crossover frequency. The loop was compensated in three steps. The OTA tail currents were reduced to lower g_m . The OTA mirror loads were made narrower to move the second pole higher. A dominant pole was set by a PMOS MOS capacitor on v_{gp} (14 fingers). The resulting phase margin is at least 68° over 0.2–0.9 V, with about 74° at the lock point (Section 4.3.3).

3.2.3 Voltage-Controlled Oscillator

Topology and governing equation. The VCO is a 7-stage single-ended current-starved ring [12] (Figure 5). Each stage is an inverter (PM9–PM15 / NM0–NM6) with a PMOS header (PM2–PM8) and an NMOS footer (NM7–NM13) that limit its charging and discharging current. A bias branch converts V_c into the bias voltages v_{bp} and v_{bn} , and a small output buffer drives the divider. A ring of N inverting stages oscillates only if N is odd, with period $T = 2Nt_d$. In a starved stage the output capacitance C_L is charged by the limited current I_D and switches at about $V_{DD}/2$:

$$t_d \approx \frac{C_L (V_{DD}/2)}{I_D} \Rightarrow f_{osc} = \frac{1}{2Nt_d} = \frac{I_D}{N C_L V_{DD}} \quad (3.10)$$

At the top of the tuning range (1.5 GHz, $N = 7$), each stage must switch in $t_d = 1/(2 \times 7 \times 1.5 \text{ GHz}) = 47.6 \text{ ps}$.

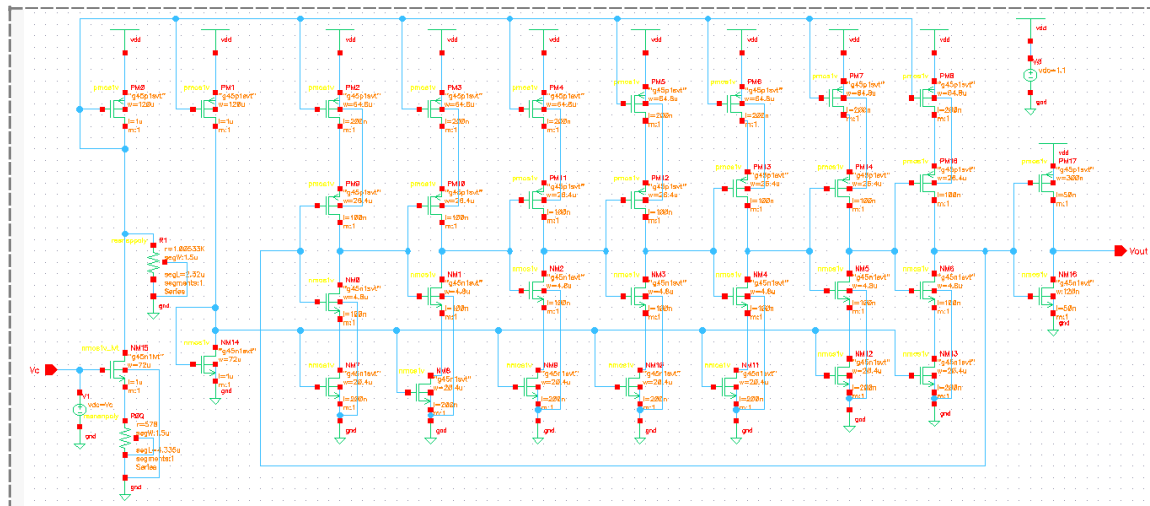


Figure 5: Schematic of the 7-stage current-starved ring VCO.

Sizing method (W/L sweeps). Each group of transistors controls one term of this equation, so each group was swept for its own purpose with ADE parametric analysis instead of changing all sizes at once:

5. *Set f_{max} .* With V_c at the top of its range, the starving widths and inverter sizes were swept until the ring ran at about 1.5 GHz. The first version free-ran at about 2.45 GHz because the inverters were too small for the current. This was corrected through the I/C ratio, not by changing the number of stages.
6. *Set f_{min} .* V_c was swept from 0.2 to 0.9 V and the frequency at 0.2 V was checked. If it was above 900 MHz, the starving current was reduced. If the ring stopped oscillating, the minimum current was increased.
7. *Match header and footer currents.* An early version had very different mirror ratios on the two sides (about 1:16 and 1:167), which gave very asymmetric rise and fall times. The ratios

were matched to give about 50 % duty cycle and symmetric edges, which also improves phase noise.

8. *Width limit.* GPD045 models are characterised up to 10 μm per finger, so every larger width was drawn with multiple fingers.

The effect of each parameter, found from these sweeps, is summarised in Table 4.

Device	Increase W	Increase L
Inverters (PM9–PM15, NM0–NM6)	More C _L and drive; C dominates, so f decreases. Sharper edges, slightly lower phase noise, more power	Much more C and less drive, so f falls steeply. Kept short (100 nm)
Starving devices (PM2–PM8, NM7–NM13)	More current per stage: f _{max} , tuning range and K _{VCO} increase. P/N ratio sets duty cycle	Less current, higher r _o ; devices stay saturated over more of the swing, giving a more linear curve and lower noise
Input NMOS NM15 (gate = V _c)	Larger g _m : K _{VCO} increases strongly, f–V curve shifts up	Smaller g _m : K _{VCO} decreases, more linear; less flicker noise
Bias / mirror (PM0, PM1, NM14)	Width ratio to the starving devices sets the mirror ratio: more current per stage, higher f and K _{VCO}	Better matching and lower flicker noise, no speed penalty (not in the signal path)
V _{th} of input device (LVT)	–	Lower V _{th} : current starts at lower V _c , tuning spreads over 0.2–0.9 V
R0 (degeneration)	–	Larger R0: lower, more linear K _{VCO} ; transistor noise reduced by $(1 + g_m R0)^2$
R1 (offset path)	–	Smaller R1: more fixed current, so f _{min} rises and the tuning span narrows
All W $\times$ k, all R $\div$ k	Same f and tuning curve; phase noise about 10 log k lower; k times the power	–

Table 4: Effect of transistor parameters on VCO properties.

In short, the inverters set where the f–V curve sits, the starving devices set how wide it is, and the bias and input devices set how steep it is (K_{VCO}) and how noisy it is.

Achieving the frequency range. A plain current-starved ring is far too sensitive. If V_c drives the starving devices directly, the current goes from zero to maximum within a few hundred millivolts, giving a K_{VCO} of roughly 8–12 GHz/V. The specification requires

$$\frac{K_{VCO}}{f_{center}} = \frac{750 \text{ MHz/V}}{1.2 \text{ GHz}} = 0.625 \text{ V}^{-1}, \quad \frac{K_{VCO}}{f_{osc}} = \frac{1}{I_D} \frac{\partial I_D}{\partial V_c} = \frac{I_{var}}{I_D} \cdot \frac{g_m}{I_{var}} \quad (3.11)$$

Since g_m/I_D of a MOSFET is at least about 2 V^{-1} , only a fraction $I_{var}/I_D \approx 0.3$ of the stage current may depend on V_c . The rest must be fixed. Two resistors in the bias branch implement this:

- *R1 (offset current path).* R1 carries a fixed current even when the input transistor is nearly off. This keeps the ring oscillating at low V_c and sets $f_{min} \approx 900 \text{ MHz}$. V_c only adds current on top of it, up to f_{max} .
- *R0 (source degeneration of the input NMOS).* The effective transconductance becomes

$$G_{m,eff} = \frac{g_m}{1 + g_m R_0}, \quad K_{VCO} = \frac{\partial f_{osc}}{\partial V_c} = \frac{1}{N C_L V_{DD}} \cdot \frac{\partial I_D}{\partial V_c} \approx \frac{M G_{m,eff}}{N C_L V_{DD}} \quad (3.12)$$

where M is the mirror ratio from the bias branch to each stage. R0 lowers the slope and makes it more linear across the range. The measured tuning range is 900.7 MHz to 1.50 GHz over $V_c = 0.2$ –0.9 V. This covers the required 0.9–1.5 GHz ($\pm 25\%$ around 1.2 GHz), with lock at $V_c \approx 0.58 \text{ V}$ (575.7 mV in the locked PLL).

Low-threshold (LVT) input transistor. With a standard NMOS ($V_{th} \approx 0.4$ V), the input transistor carries almost no current below about 0.45 V. All the tuning would be squeezed into 0.45–0.9 V, making K_{VCO} much steeper and leaving part of the charge pump’s matched window unused. The 0.2–0.9 V window was specified because the charge pump is matched there, so the VCO has to use all of it. A low- V_{th} device (g45n1lvt) starts conducting at a lower voltage and spreads the tuning over the full window, which gives the lowest possible average K_{VCO} for the required range.

Phase noise reduction. The first working design gave -76.8 dBc/Hz at 1 MHz offset, against a -90 dBc/Hz target. Two noise mechanisms were reduced one at a time:

- *Flicker noise (reduced with area).* Flicker noise comes from carrier trapping at the gate oxide, and its input-referred power falls with gate area [1], $\overline{v_n^2} = \frac{K_f}{C_{ox}WL} \cdot \frac{1}{f}$. Noise in the bias current maps directly into frequency noise. The bias and mirror devices, which are not in the signal path, were therefore lengthened from 45 nm to 1 μ m, while the ring inverters stayed short. The phase noise improved to -82 dBc/Hz and the flicker corner moved from above 1 MHz down to 255 kHz.
- *Thermal noise (reduced with power).* With the corner below the 1 MHz spec point, the noise there was thermal, and the noise summary showed no dominant device (the largest contributor was 13.7 %). Thermal phase noise of a ring oscillator falls inversely with power [13], [14], $\mathcal{L}(\Delta f) \propto \frac{kT}{P} \left(\frac{f_0}{\Delta f} \right)^2$. All widths were therefore multiplied by $k = 12$ and all bias resistors divided by 12, with the lengths unchanged. Every current and capacitance scales by k , so $f_{osc} = I_D / (NC_L V_{DD})$ and the whole tuning curve stay the same, while the noise improves by about $10 \log_{10} 12 = 10.8$ dB.

After the scaling step the VCO reached -92.8 dBc/Hz with a flicker corner of 45.7 kHz. The final VCO, as integrated in the PLL, achieves -91.9 dBc/Hz at 1 MHz offset, 1.9 dB better than the specification. Its power rose from about 150 μ W to about 1.8 mW, still well within the 5 mW budget. Degeneration also helps here, because it suppresses the input transistor’s noise current by $(1 + g_m R_0)^2$.

K_VCO. The tuning range must cover 600 MHz inside the 0.7 V window where the charge pump is matched, so the average gain must be at least

$$\bar{K}_{VCO} \geq \frac{1.5-0.9 \text{ GHz}}{0.9-0.2 \text{ V}} = 857 \text{ MHz/V} > 750 \text{ MHz/V} \quad (3.13)$$

The two specifications cannot both hold on average, so 750 MHz/V can only be a local slope at the lock point. K_{VCO} was measured as the derivative of the $f-V_c$ curve at 1.2 GHz, giving about 1.08 GHz/V. Lowering it further would have shrunk the tuning range below 0.9–1.5 GHz, which is a hard requirement. The loop uses K_{VCO} only through the product $I_{CP} K_{VCO} / C$, so the loop filter was instead recalculated for the measured value, as shown in the next subsection.

Output buffer. The inverter at the ring output (PM17/NM16) isolates the last ring node from the divider’s input capacitance, so all stages see an equal load and the frequency is not shifted. It also sharpens the edges for the TSPC divider. The buffer was not scaled with the ring. It was kept small (PMOS 300 nm / 50 nm, NMOS 120 nm / 50 nm) so that it adds very little load to the last ring node. The final VCO device sizes are listed in Table 5.

Device(s)	Role	W / L
NM15 (LVT NMOS, gate = V_c)	V-to-I input device, sets K_VCO	72 μ m / 1 μ m

Device(s)	Role	W / L
R0	Source degeneration: lowers and linearises K_{VCO}	578 Ω
R1	Offset-current path: sets f_{min}	1.005 k Ω
PM0 (diode), PM1	PMOS bias and mirror	120 μm / 1 μm
NM14 (diode)	NMOS bias, generates v_{bn}	72 μm / 1 μm
PM2–PM8	PMOS starving (header) devices	64.8 μm / 200 nm
NM7–NM13	NMOS starving (footer) devices	20.4 μm / 200 nm
PM9–PM15	Ring inverter PMOS (7 stages)	26.4 μm / 100 nm
NM0–NM6	Ring inverter NMOS (7 stages)	4.8 μm / 100 nm
PM17 / NM16	Output buffer	300nm / 50nm 120nm / 50nm

Table 5: Final VCO device sizes.

3.2.4 Loop Filter

Linear phase-domain model. Near lock, every block can be replaced by its small-signal phase gain (Figure 6): $I_{CP}/2\pi$ for the PFD and charge pump, the impedance $Z_{LF}(s)$ for the filter, $2\pi K_{VCO}/s$ for the VCO (which integrates frequency into phase), and $1/N$ for the divider.

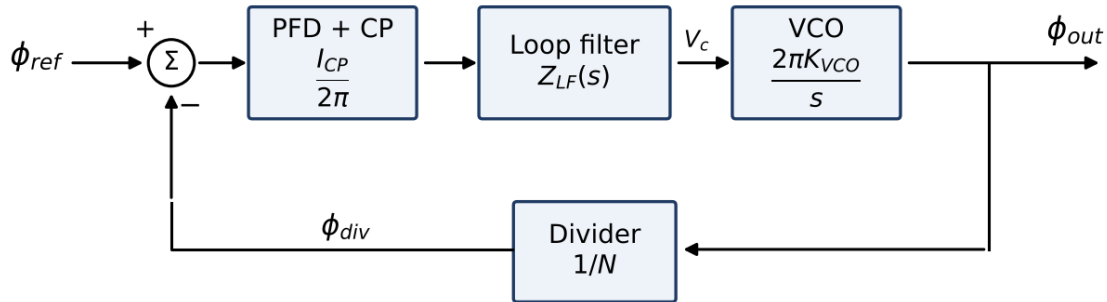


Figure 6: Linear phase-domain model of the charge-pump PLL used for loop design.

The required filter is a series R–C1 branch in parallel with C2:

$$Z_{LF}(s) = \left(R + \frac{1}{sC_1} \right) \parallel \frac{1}{sC_2} = \frac{1+sRC_1}{s(C_1+C_2)(1+sR\frac{C_1C_2}{C_1+C_2})}, \quad \omega_z = \frac{1}{RC_1}, \quad \omega_p = \frac{C_1+C_2}{RC_1C_2} \quad (3.14)$$

With K_{VCO} the VCO gain in Hz/V, the VCO adds a phase gain of $2\pi K_{VCO}/s$ and the open-loop gain is

$$L(s) = \frac{I_{CP}}{2\pi} \cdot Z_{LF}(s) \cdot \frac{2\pi K_{VCO}}{s} \cdot \frac{1}{N} = \frac{I_{CP}K_{VCO}}{N(C_1+C_2)} \cdot \frac{1+s/\omega_z}{s^2(1+s/\omega_p)} \quad (3.15)$$

The two poles at the origin make the loop type-II, so the static phase error is zero in lock. The pole ω_p makes it third-order, as the architecture requires. At crossover the two integrators contribute -180° , so the phase margin is

$$PM = \tan^{-1}\left(\frac{\omega_c}{\omega_z}\right) - \tan^{-1}\left(\frac{\omega_c}{\omega_p}\right) \quad (3.16)$$

The continuous-time model is valid because $f_{ref}/f_c = 75/3.75 = 20$, above the usual limit of about

10 [2]. The PFD's sampling then costs only about 1–2° of phase margin.

Design procedure [9], [10]. The targets were $f_c = 3.75$ MHz ($\omega_c = 2.356 \times 10^7$ rad/s), PM = 60°, $I_{CP} = 50$ μ A, $K_{VCO} = 750$ MHz/V and N = 16.

Step 1 – third pole. The pole suppresses ripple on V_c but costs phase. It was placed at $f_p = 5f_c = 18.75$ MHz, which costs $\tan^{-1}(1/5) = 11.3^\circ$.

Step 2 – zero. The zero must therefore supply $60^\circ + 11.3^\circ = 71.3^\circ$:

$$\frac{\omega_c}{\omega_z} = \tan(71.3^\circ) = 2.956 \Rightarrow f_z = \frac{3.75 \text{ MHz}}{2.956} = 1.27 \text{ MHz} \quad (3.17)$$

Step 3 – capacitor ratio.

$$\frac{\omega_p}{\omega_z} = \frac{C_1 + C_2}{C_2} = 5 \times 2.956 = 14.78 \Rightarrow \frac{C_1}{C_2} = 13.78 \quad (3.18)$$

Step 4 – total capacitance from $|L(j\omega_c)| = 1$:

$$C_1 + C_2 = \frac{I_{CP} K_{VCO}}{N \omega_c^2} \cdot \frac{|1 + j\omega_c/\omega_z|}{|1 + j\omega_c/\omega_p|} = \frac{2343.75 \times 3.121}{5.552 \times 10^{14} \times 1.020} = 12.92 \text{ pF} \quad (3.19)$$

Step 5 – split and find R.

$$C_2 = \frac{12.92 \text{ pF}}{14.78} = 0.874 \text{ pF}, \quad C_1 = 12.05 \text{ pF}, \quad R = \frac{1}{\omega_z C_1} = 10.4 \text{ k}\Omega \quad (3.20)$$

These values give a crossover of 3.750 MHz and a phase margin of 60.0°.

Rescaling for the measured VCO gain. The transistor-level VCO has $K_{VCO} \approx 1.08$ GHz/V at the lock point, $k = 1.08/0.75 = 1.41$ times the design value (see the VCO subsection). The loop gain is proportional to $K_{VCO}/(C_1 + C_2)$, so the crossover is restored by multiplying both capacitors by k and dividing R by k . The products RC_1 and RC_2 are unchanged, so the zero, the pole and the phase margin stay in place:

$$R' = \frac{R}{k} = 7.37 \text{ k}\Omega, \quad C_1' = kC_1 = 17.02 \text{ pF}, \quad C_2' = kC_2 = 1.235 \text{ pF} \quad (3.21)$$

PDK implementation. For layout and LVS, the filter was built (Figure 7) from a resnspoly poly resistor and two arrays of mimcap unit cells. The capacitance comes in unit-cell steps, and the poly resistor has an extra end/contact resistance of about 0.68 k Ω that is not shown in its CDF value. The final VCO measurement gives a gain of about 1.08 GHz/V, about 2 % above the value used here. The drawn values were therefore trimmed in the linear model, with the measured VCO gain, until it read 3.752 MHz and 60.01° (Table 6).

Component	Nominal design (K_VCO = 750 MHz/V)	Rescaled (K_VCO ≈ 1.06 GHz/V)	PDK implementation
R	10.4 k Ω	7.37 k Ω	resnspoly, segL = 10.5 μ m $\rightarrow$ 6.82 k Ω (+ ≈ 0.68 k Ω end resistance)
C1, series (C0 in schematic)	12.05 pF	17.02 pF	mimcap, 37.54 fF $\times$ m = 445 $\rightarrow$ 16.71 pF
C2, shunt (C1 in schematic)	0.874 pF	1.235 pF	mimcap, 36.67 fF $\times$ m = 33 $\rightarrow$ 1.21 pF
f_z / f_p	1.27 MHz / 18.75 MHz	1.27 MHz / 18.75 MHz	trimmed to the same
f_c / PM	3.75 MHz / 60.0°	3.75 MHz / 60.0°	3.752 MHz / 60.01° (linear model)

Table 6: Loop filter component values.

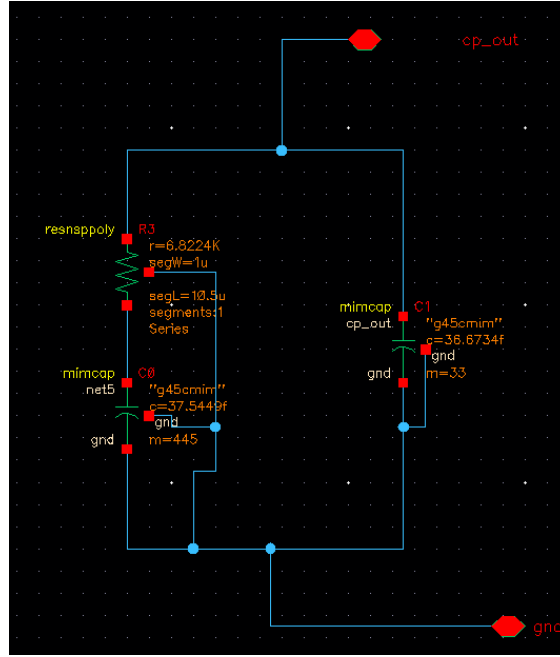


Figure 7: Loop filter schematic with GPDK045 devices: resnappoly R3 (6.82 k Ω), mimcap C0 ($m = 445$, 16.71 pF) and mimcap C1 ($m = 33$, 1.21 pF).

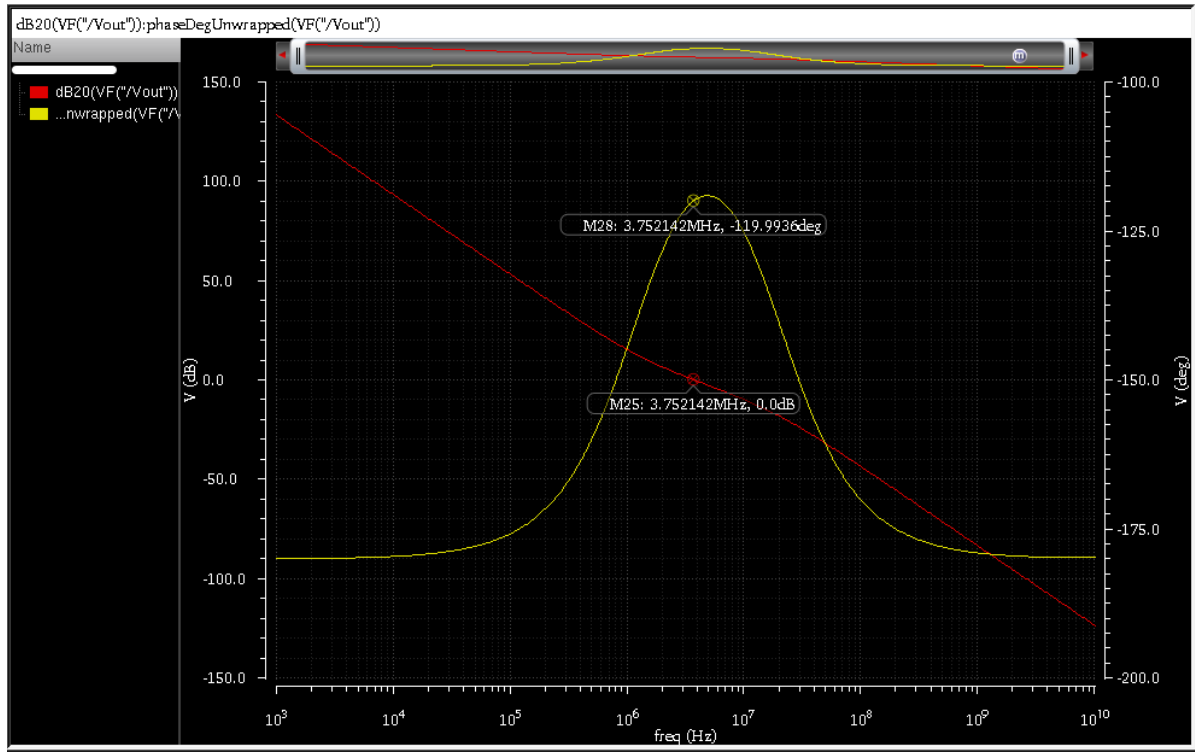


Figure 8: Calculated open-loop Bode plot with the rescaled loop filter. The phase peak lies close to crossover, which keeps the phase margin insensitive to changes in loop gain.

Lock-time estimate. Neglecting C_2 , the loop is a second-order system with

$$\omega_n = \sqrt{\frac{I_{CP} K_{VCO}}{N C_1}} = 1.40 \times 10^7 \text{ rad/s}, \quad \zeta = \frac{R C_1}{2} \omega_n = 0.88 \quad (3.22)$$

The linear settling time is $t_s \approx 4/(\zeta \omega_n) \approx 0.33 \mu\text{s}$. During acquisition, V_c cannot change faster than $I_{CP}/(C_1 + C_2) \approx 2.7 \text{ V}/\mu\text{s}$, so reaching the lock point (about 0.58 V) takes at least 0.2 μs . Adding the linear settling time gives a minimum lock time of roughly 0.5 μs . Cycle slipping during acquisition

can lengthen this to a few microseconds, but either way it is well inside the 8 μ s limit.

Ripple filtering. At the reference frequency, the 18.75 MHz pole gives an additional $20\log_{10}|1 + j 75/18.75| \approx 12.3$ dB of attenuation of charge-pump ripple on V_C , which helps meet the -40 dBc reference spur specification.

3.2.5 Frequency Divider

Since $N = 16 = 2^4$, the divider is a ripple chain of four positive-edge TSPC D flip-flops, each wired as a toggle ($\bar{Q} \rightarrow D$), giving $1.2 \text{ GHz} \rightarrow 600 \rightarrow 300 \rightarrow 150 \rightarrow 75 \text{ MHz}$ (Figure 9), and each stage is the TSPC flip-flop of Figure 10. TSPC (true single-phase clock) logic [8] was chosen because it needs only one clock phase and about nine transistors per flip-flop. It therefore loads the VCO less and runs faster than a transmission-gate master–slave flip-flop.

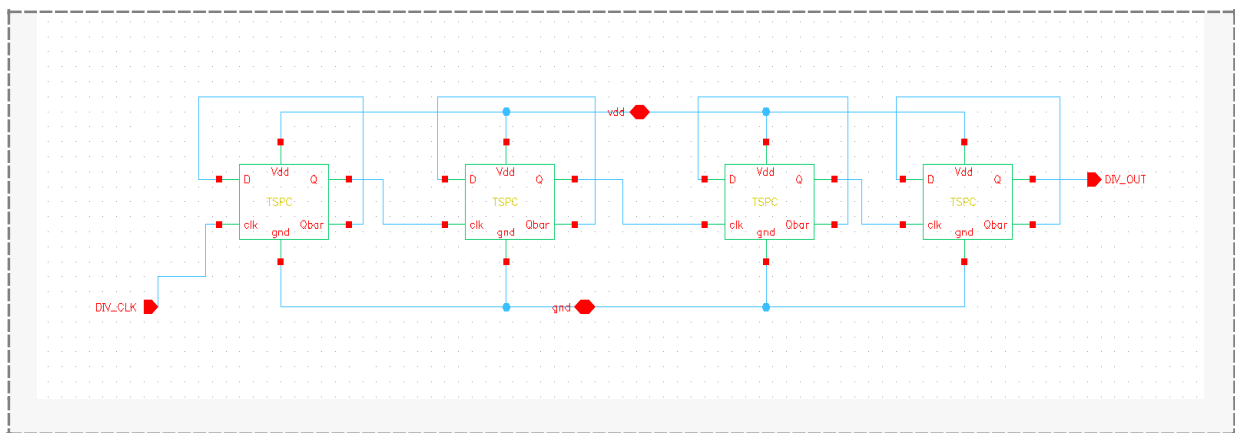


Figure 9: The four-stage divide-by-16 ripple chain.

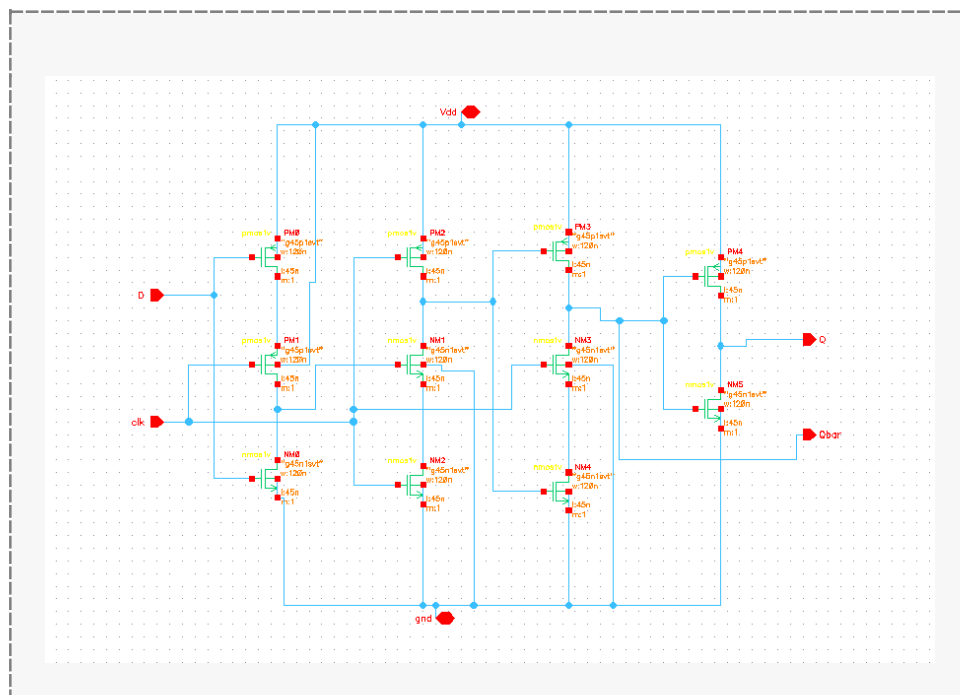


Figure 10: TSPC D flip-flop.

A toggle stage works if its output change can travel back through the feedback path before the next

clock edge:

$$t_{clk \rightarrow Q} + t_{setup} < T_{in,min} = \frac{1}{1.5 \text{ GHz}} = 667 \text{ ps} \quad (3.23)$$

This is critical only for the first stage, since each later stage runs at half the rate of the one before. The accumulated ripple delay ($4t_{clk \rightarrow Q}$) only adds a constant phase offset at the PFD input, which the type-II loop removes. The dynamic power of each stage is proportional to its clock frequency, so

$$P_{div} \approx C V_{DD}^2 f_{in} \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) = 1.875 C V_{DD}^2 f_{in} \quad (3.24)$$

and the first stage dominates. TSPC nodes are dynamic and need a minimum clock rate to hold their charge against leakage. The slowest stage runs at 150 MHz, far above that limit.

3.2.6 Summary of Design Values

Table 7 collects the main design equations of this section and the values obtained from them.

Quantity	Design equation	Value
Division ratio	$N = f_{out} / f_{ref}$	16 ($4 \times \div 2$)
PFD gain	$I_{CP} / 2\pi$	7.96 $\mu\text{A/rad}$ (0.05 fC/ps)
PFD minimum pulse	$t_{reset} > t_{on,CP}$	$\approx 225 \text{ ps}$
CP r_o needed (simple mirror)	$r_o \geq 2\Delta V / \Delta I_{max}$	$\geq 280 \text{ k}\Omega$
CP reference (beta multiplier, $K = 4$)	$I = 1 / (2\mu_{nC_{ox}}(W/L)R_{S^2})$	51.1 μA
VCO stage delay at 1.5 GHz	$t_d = 1 / (2N f)$	47.6 ps
Minimum average K_{VCO}	$\Delta f / \Delta V_c$	857 MHz/V
VCO noise scaling	$10 \log_{10} k, k = 12$	$\approx 10.8 \text{ dB}$
Loop zero / pole	$\omega_z = 1/RC1, \omega_p = (C1+C2)/RC1C2$	1.27 MHz / 18.75 MHz
Crossover / phase margin	$ L(j\omega_c) = 1; PM = \tan^{-1}(\omega_c/\omega_z) - \tan^{-1}(\omega_c/\omega_p)$	3.75 MHz / 60°
Natural frequency / damping	$\omega_n = \sqrt{(I_{CP} K_{VCO} / N C1)}, \zeta = RC1\omega_n / 2$	2.22 MHz / 0.88
Settling / slew rate	$t_s \approx 4 / \zeta\omega_n; I_{CP} / (C1+C2)$	$\approx 0.33 \mu\text{s} / \approx 2.7 \text{ V}/\mu\text{s}$
Divider critical timing	$t_{clk \rightarrow Q} + t_{setup} < 1 / f_{max}$	$< 667 \text{ ps}$

Table 7: Summary of the main calculated design quantities.

3.2.7 Layout Design

The layout was drawn in Cadence Virtuoso Layout XL following the course layout guidelines [5]. Each block was laid out separately and then placed in the top-level cell PLL_TOP. For every block, the device layouts were generated directly from the finished schematic with “Generate All From Source”, which places the GPDK045 PCells, MOSFETs, the resnspoly resistor and the mimcap capacitors, with exactly the sizes, finger counts and multipliers used in simulation. All interconnect between the devices was then routed by hand. A floorplan was made before routing to give each block its own area, and the blocks were arranged into a compact, near-rectangular outline of $933.09 \mu\text{m} \times 669.89 \mu\text{m}$ (0.6251 mm^2). The loop-filter mimcap array ($m = 445$ and $m = 33$ unit cells) takes up most of this area, about 89 % of the combined block area, so it was placed first and the smaller blocks were arranged around it.

Use of different metal layers. Several metal layers were used for two reasons. The first is set by the process and the guidelines. Metal1 was used mainly for short connections inside and between

neighbouring devices, and Metal2 and Metal3 for longer signal routes between devices and between blocks. In addition, the mimcap is formed between Metal10 and Metal11, so its two terminals lie on these upper layers by construction. The second reason is to avoid overlapping. Two wires on the same metal layer cannot cross or overlap, because they would be shorted together. Wherever two nets had to cross, one of them was moved to a different metal layer, so the two wires pass over each other on separate levels, isolated by the dielectric between them.

Use of vias. A via is a vertical metal plug through the dielectric that electrically joins two adjacent metal layers. Every change of layer in the routing therefore needed a via: VIA1 between Metal1 and Metal2, VIA2 between Metal2 and Metal3, and so on up to VIA10 between Metal10 and Metal11, while contacts connect the polysilicon gates and the diffusion regions to Metal1. A typical crossing therefore uses a via on each side, taking one net up to the next metal, across the other wire, and back down. The mimcap terminals on Metal10 and Metal11 were likewise brought down to the routing layers through stacks of vias, as the guideline requires. All vias were placed with the metal enclosure required by the GPDK045 rules [3]. Figure 11 shows a via stack between Metal1 and Metal11.

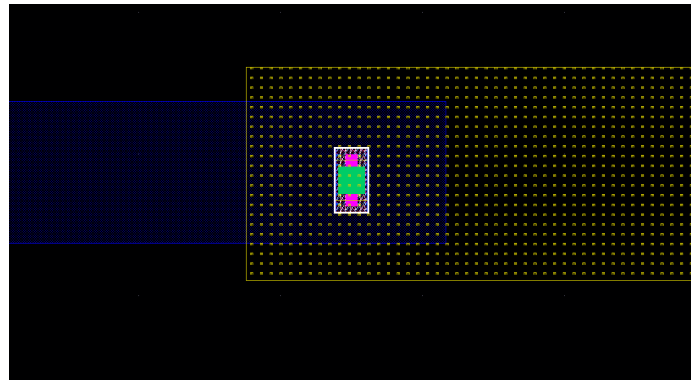


Figure 11: Via contact between Metal1 and Metal11.

3.3 Circuit Diagram

Figure 12 shows the complete transistor-level PLL assembled in the top-level cell PLL_TOP.

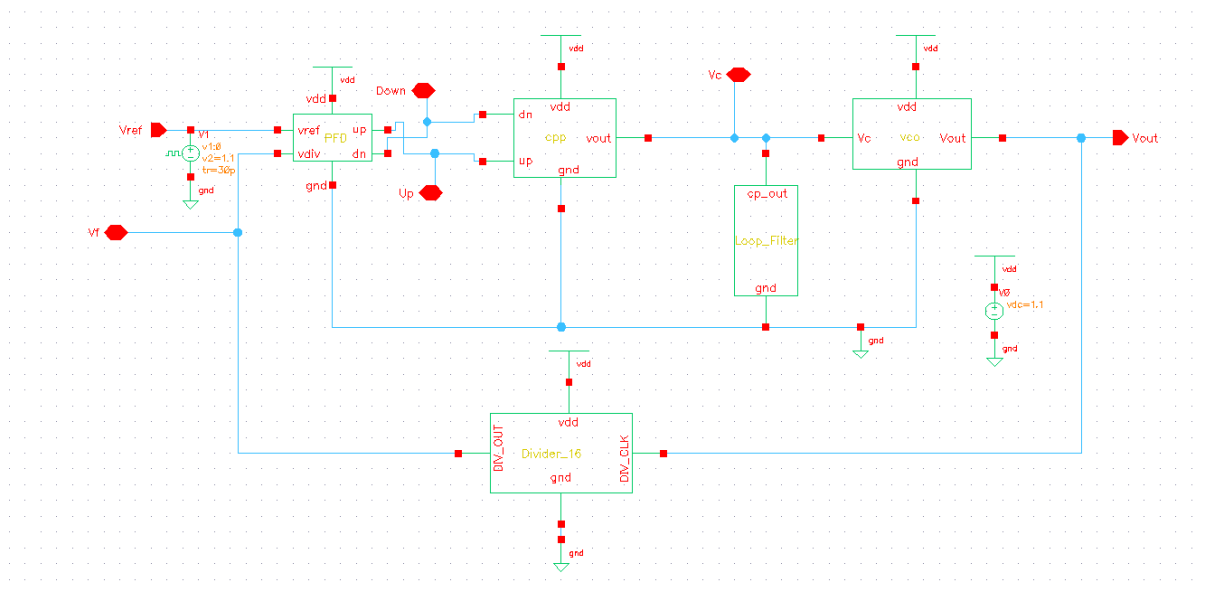


Figure 12: Complete PLL circuit diagram.

3.4 Simulation Model

Figure 13 shows the linear loop model used for the AC analysis of the loop bandwidth and phase margin (Section 4.3).

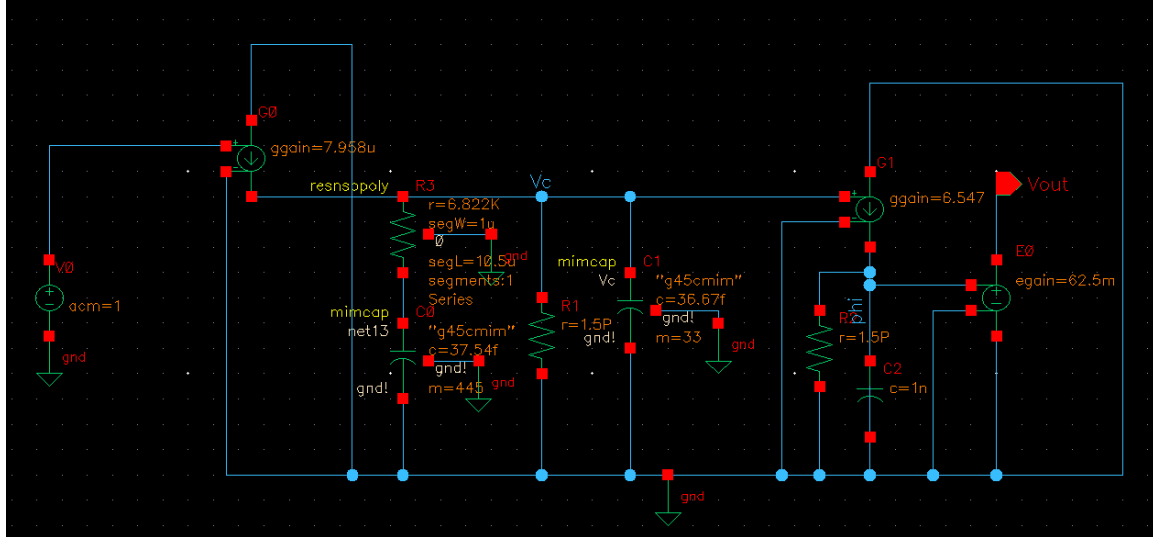


Figure 13: Linear loop model used for AC analysis.

3.5 CAD/Hardware Design (Layout)

Figures 14–25 show the layout of each block and of the complete PLL, each followed by its clean DRC and LVS result.

3.5.1 PFD Block

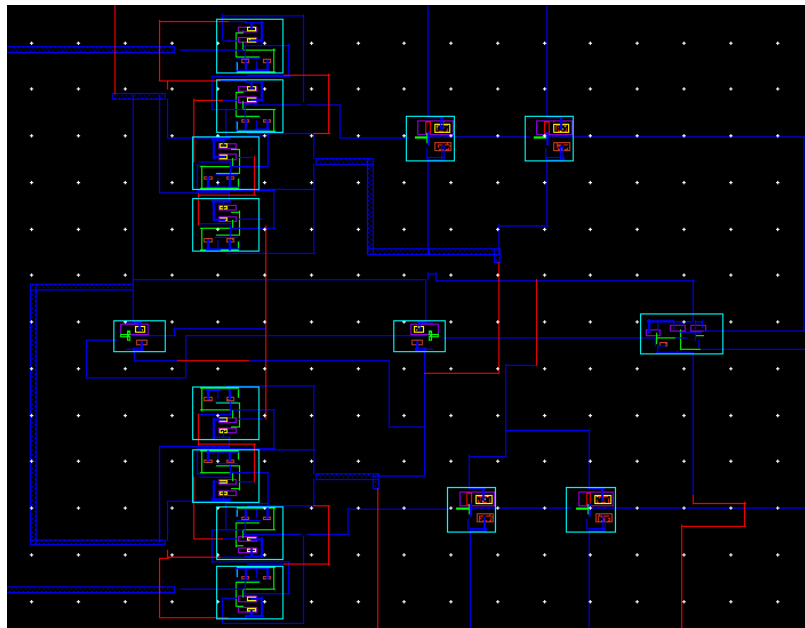


Figure 14: Layout of the PFD.

Width – 109 μm
Height – 99.395 μm

Area = 10,834 μm^2
= 0.01083 mm^2

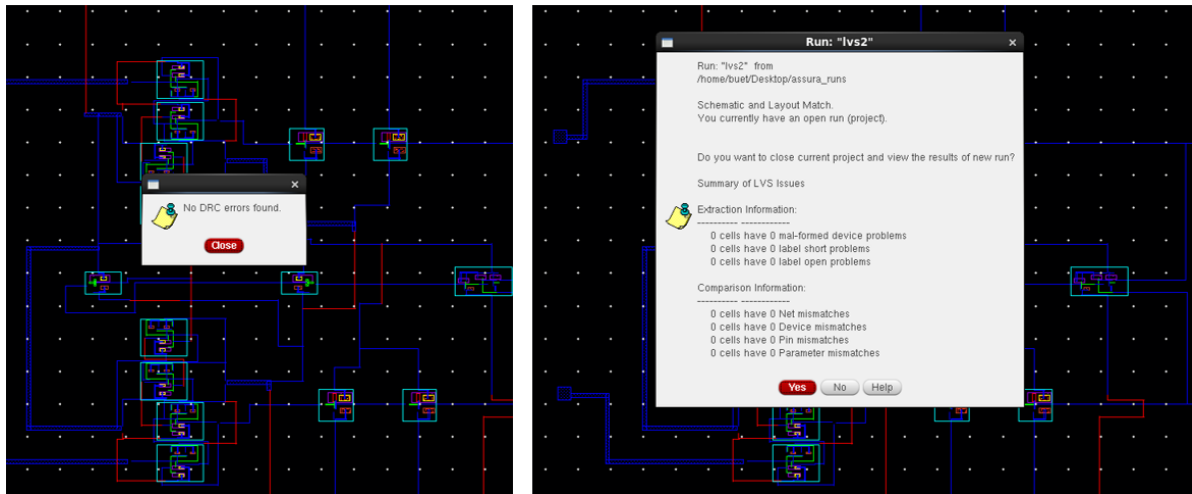


Figure 15: PFD layout: DRC and LVS clean.

3.5.2 Charge Pump Block

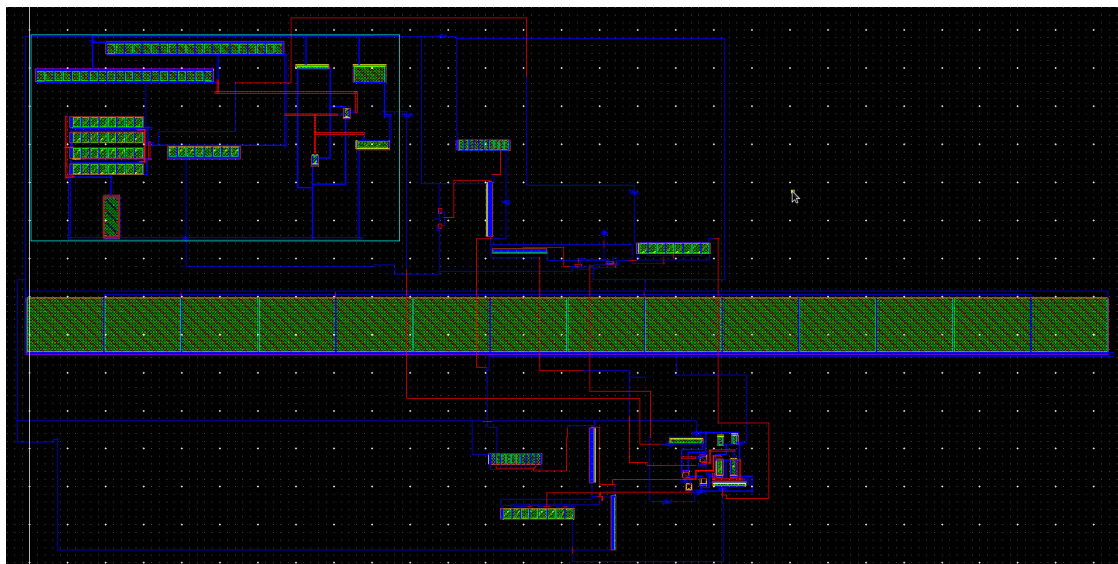


Figure 16: Layout of the charge pump.

Width – 144.21 μm
Height - 71.32 μm
Area = 10,285 μm^2 = 0.01029 mm^2

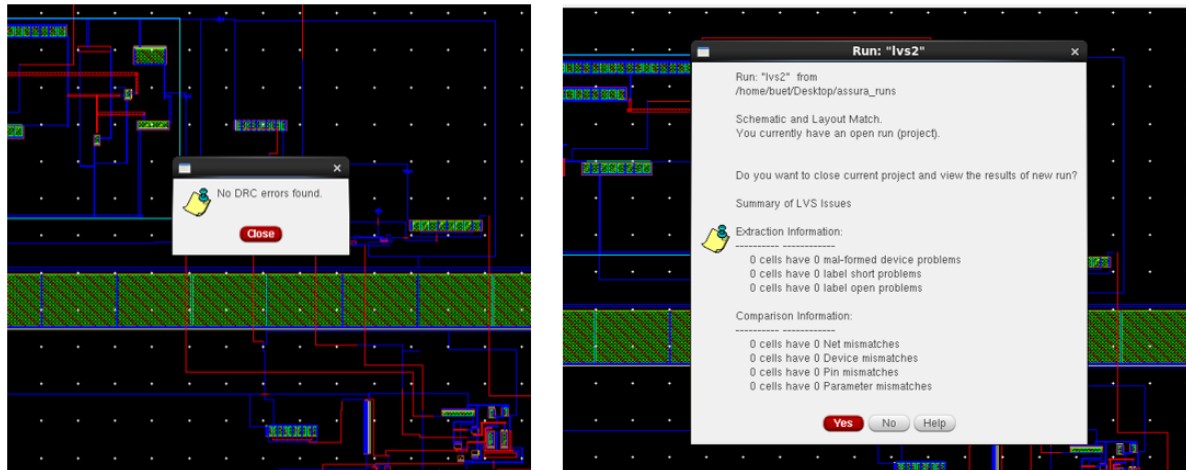


Figure 17: Charge-pump layout: DRC and LVS clean.

3.5.3 VCO Block

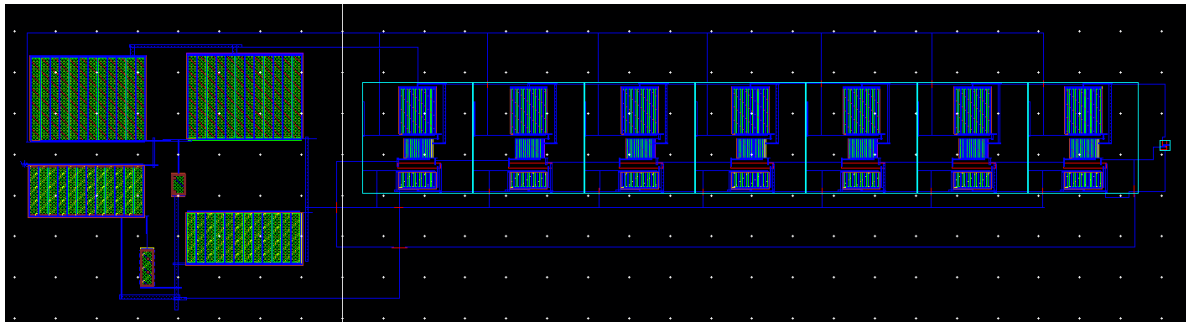


Figure 18: Layout of the VCO.

Width – 139.865 μm
 Height – 33.825 μm
 Area = 4,731 μm^2 = 0.004731 mm^2

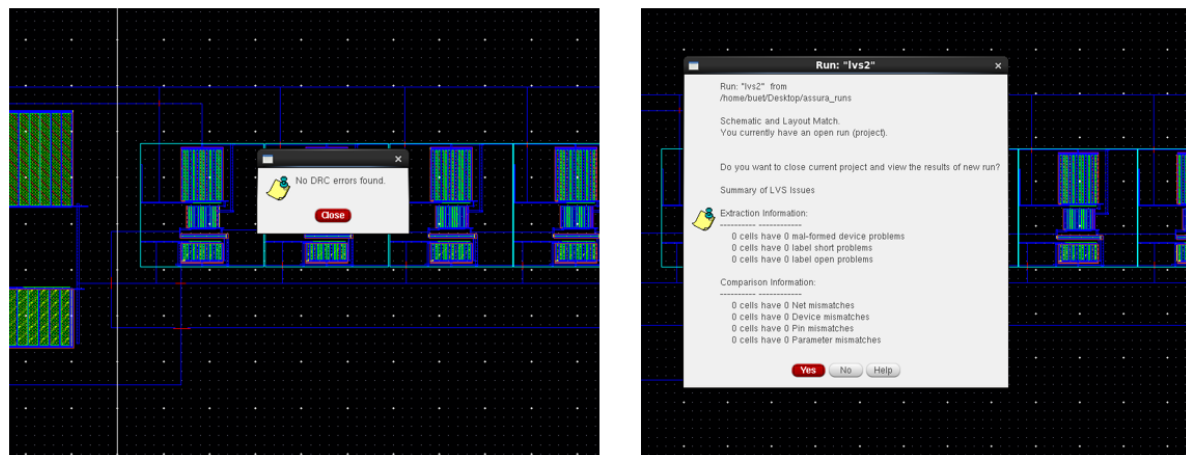


Figure 19: VCO layout: DRC and LVS clean.

3.5.4 Loop Filter Block

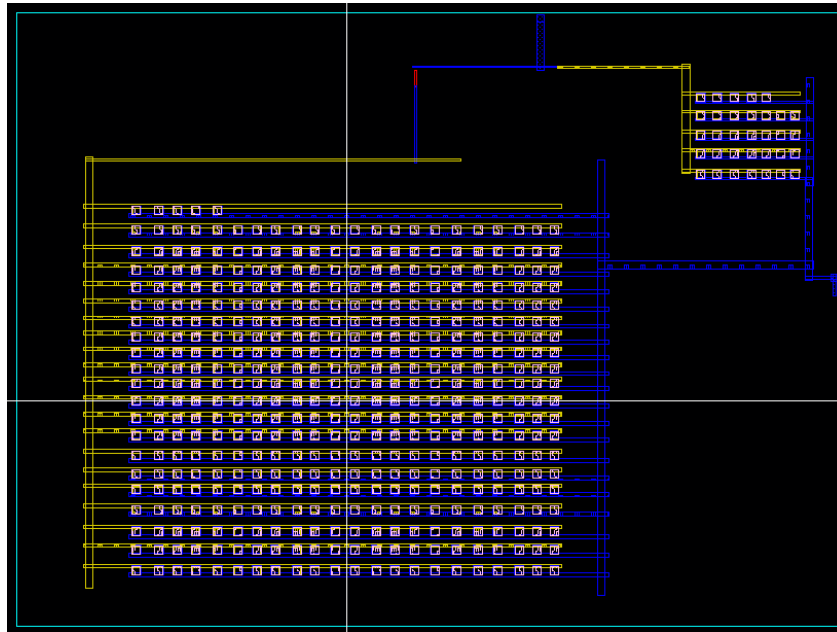


Figure 20: Layout of the loop filter.

Width – 513.135 μm
Height – 394.495 μm
Area = 202,429 μm^2
= 0.2024 mm^2

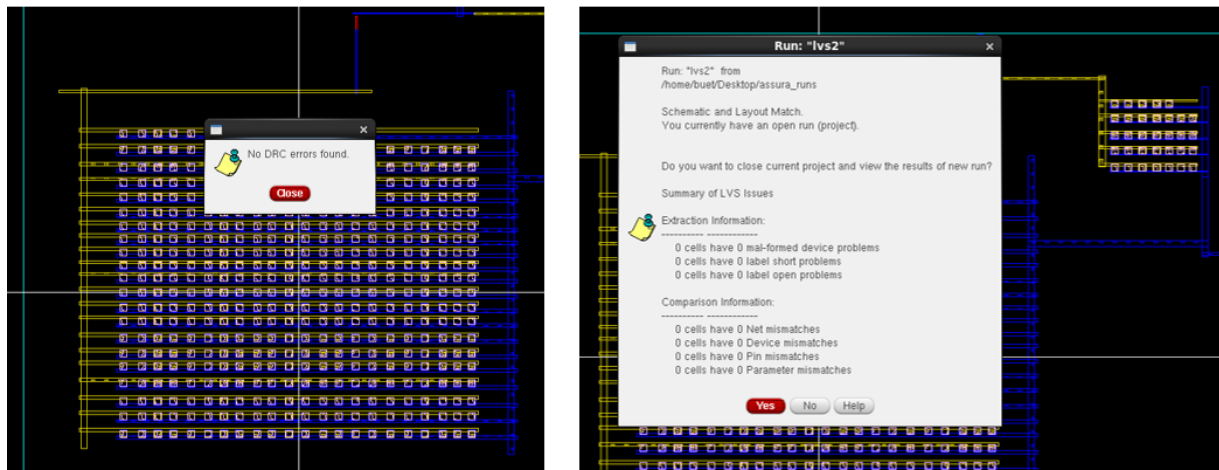


Figure 21: Loop-filter layout: DRC and LVS clean.

3.5.5 Frequency Divider Block

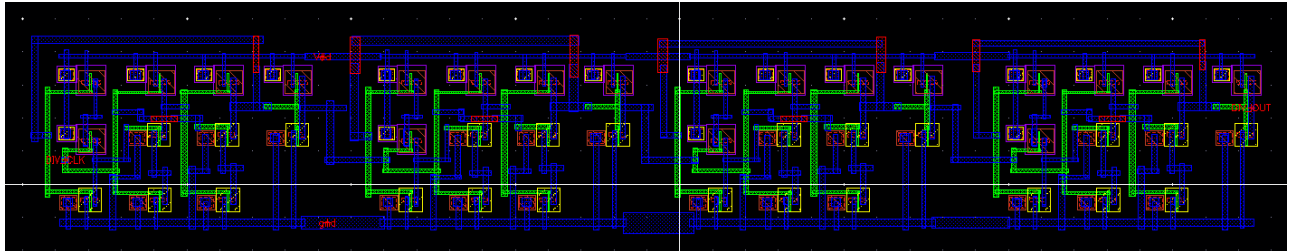


Figure 22: Layout of the frequency divider.

Width – 37.405 μm
Height – 5.7 μm
Area = 213.2 μm^2
= 0.0002132 mm^2

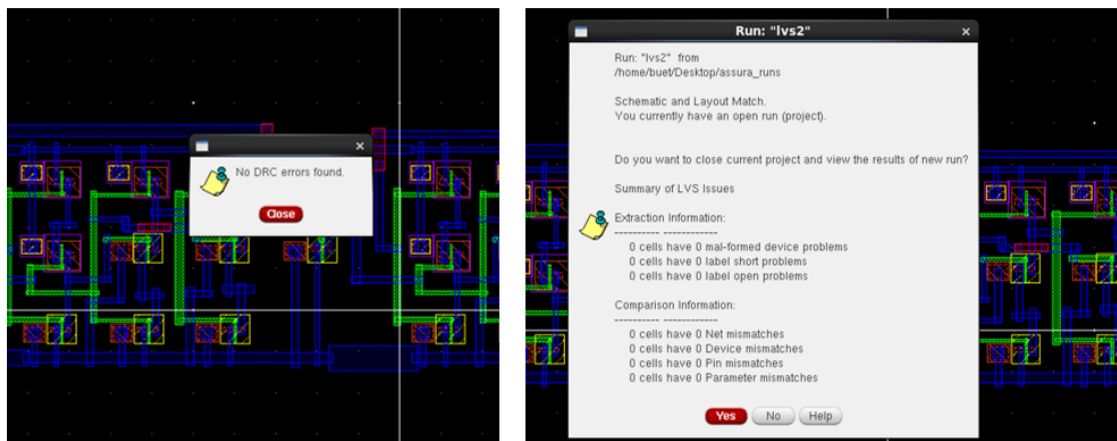


Figure 23: Divider layout: DRC and LVS clean.

3.5.6 PLL Block

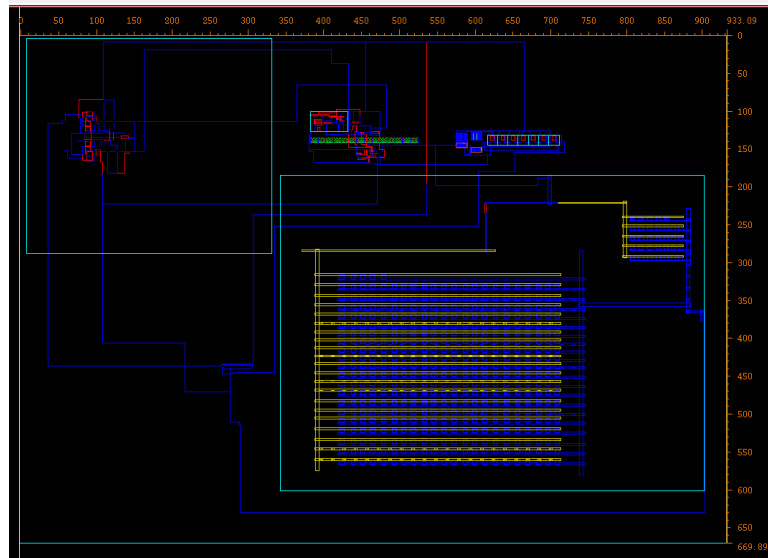


Figure 24: Layout of the complete PLL.

Width – 933.09 μm
Height – 669.89 μm
Area = 625,068 μm^2
= 0.6251 mm^2

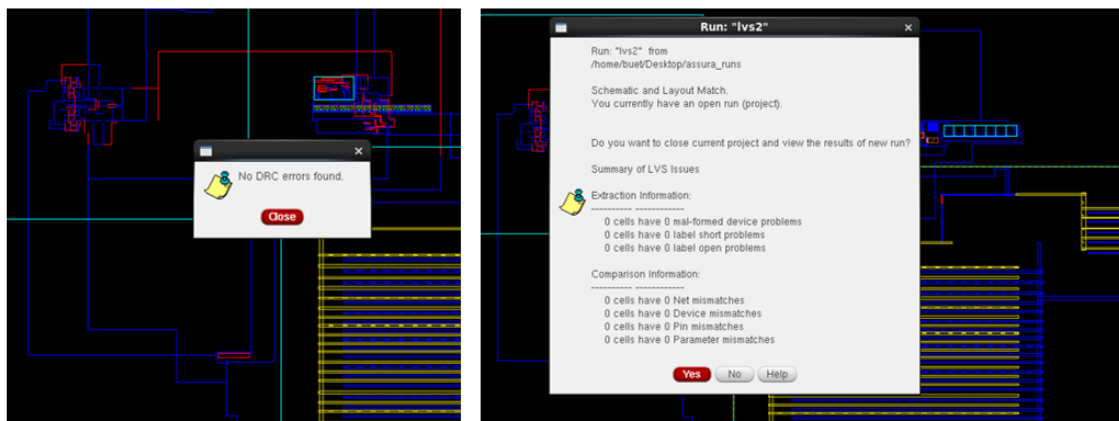


Figure 25: Complete PLL layout: DRC and LVS clean.

4 Design Analysis and Evaluation

4.1 Novelty

The charge-pump PLL is a mature architecture, and none of its blocks is a new circuit in itself. The novelty of this work lies in how a partly over constrained specification was resolved at transistor level in a 45 nm process at 1.1 V:

1. **Loop filter designed last, from the measured K_{VCO} .** The specified K_{VCO} (750 MHz/V) and tuning range (0.9–1.5 GHz over 0.2–0.9 V, needing ≥ 857 MHz/V) cannot both be met. The tuning range was kept, K_{VCO} was measured (≈ 1.08 GHz/V), and R , C_1 and C_2 were rescaled by $k = 1.41$, preserving the zero, pole and phase margin (3.752 MHz, 60.01°). This step did

not come from a published procedure.

2. **Rail-to-rail replica-bias charge pump at 1.1 V.** With no headroom for cascoding, a replica branch with a complementary two-OTA error amplifier and an on-chip beta-multiplier reference cut the mismatch from $8.1\ \mu\text{A}$ to $+0.77 / -1.27\ \mu\text{A}$. The replica loop was found to be least stable at mid-range, the lock point, and was compensated for that worst case ($\text{PM} \geq 68^\circ$ over $0.2\text{--}0.9\ \text{V}$).
3. **VCO tuning shaped to the matched window.** An LVT input device, source degeneration (R_0) and an offset-current path (R_1) spread $900.7\ \text{MHz}\text{--}1.50\ \text{GHz}$ across exactly the $0.2\text{--}0.9\ \text{V}$ window in which the charge pump is matched.
4. **Phase noise reduced by mechanism.** Flicker noise was cut first by enlarging only the bias devices (no power cost), then thermal noise by impedance scaling ($k = 12$), which gave the predicted $10.8\ \text{dB}$ without moving the tuning curve ($-76.8 \rightarrow -91.9\ \text{dBc/Hz}$).
5. **Minimum-penalty dead-zone fix.** Two $L = 300\ \text{nm}$ inverters and output buffers give full-swing $225\ \text{ps}$ pulses, keeping the UP/DN overlap, and hence the spur, small.

4.2 Design Considerations (PO(c))

4.2.1 Considerations to public health and safety

A clock failure or excessive jitter can make a processor, communication link or medical or automotive controller corrupt data or stop responding. Reliable operation was therefore the main safety goal:

- **Safe operating levels:** $1.1\ \text{V}$ supply and $2.315\ \text{mW}$, with all devices used at their rated voltage and within the model limits, so there is no shock, heating or over-stress hazard.
- **Guaranteed start-up and lock:** a start-up circuit prevents the reference from sticking at zero current, and the frequency-detecting PFD ensures lock from any start ($612.5\ \text{ns}$ vs the $8\ \mu\text{s}$ limit).
- **Stability and timing integrity:** the replica loop is stable over the whole V_c range, the PLL has 60° phase margin, and the dead-zone-free PFD gives $1.214\ \text{ps RMS}$ jitter (limit $8\ \text{ps}$).
- **Low interference:** a reference spur of $-69.4\ \text{dBc}$ (limit $-40\ \text{dBc}$) limits interference with nearby electronic equipment.

4.2.2 Considerations to environment

- **Low power:** power was spent mainly on the VCO, where it buys phase noise; the passive filter, TSPC divider and $0.25\ \mu\text{A}$ OTA tails keep the rest small. Total power is $2.315\ \text{mW}$, 46% of the budget, about $23.5\ \text{MWh}$ per year less than the $5\ \text{mW}$ limit for every million units in continuous use.
- **Fast lock** ($0.61\ \mu\text{s}$) allows a system to power the PLL down when idle.
- **Small area:** a ring VCO (no inductor) and shortening the charge-pump sources from $4\ \mu\text{m}$ to $1\ \mu\text{m}$ reduced silicon use; the layout occupies $0.6251\ \text{mm}^2$.

4.2.3 Considerations to cultural and societal needs

- **Affordability:** a standard CMOS process with the reference current and loop filter on chip allows low-cost integration, which matters in price-sensitive markets such as Bangladesh.
- **End-user needs first:** jitter, spurs, lock time and power were treated as the binding requirements, and the K_{VCO} deviation was reported openly.
- **Local skills:** the project builds analog IC design experience with an industry-standard flow, a field in which local expertise is still limited.
- **Neutral and reproducible:** the circuit handles no personal data and is culturally neutral, and the report documents failures and fixes so others can reproduce the work.

4.3 Investigations (PO(d))

4.3.1 Design of Experiment

The design was verified bottom-up. Each block was first tested on its own, in a dedicated testbench and against its own specification, so that a fault could be traced to one block before integration. The blocks were then connected in the top-level schematic (PLL_TOP) and the full PLL was tested against the system specifications. All simulations used Cadence Spectre with the GPDK045 typical-corner models at $V_{DD} = 1.1$ V. The reference clock was an ideal 75 MHz vpulse source, as the design rules permit. Table 8 lists every experiment, the analysis used, and the quantity measured.

Block	Experiment / analysis	Stimulus or swept variable	Measured quantity	Target
PFD + CP	Transient, PFD driving the charge pump	REF and DIV edges with zero phase difference	UP/DN pulse width and swing	Full swing, no runt pulses
PFD + CP	Transient, phase-error sweep	Phase difference Δt swept through zero	Average charge per reference cycle vs Δt	Linear, no flat region (no dead zone)
CP	DC sweep (+ parametric L sweep)	$V_{out} = 0.2\text{--}0.9$ V, both switches on	I_{UP} , I_{DN} , mismatch	$50\text{ }\mu\text{A}$, $ \Delta I \leq 2.5\text{ }\mu\text{A}$
CP	DC operating point	V_{out} fixed	Current and V_{DS} of each mirror device	Diagnose mismatch
CP	Transient, separate pulses	UP pulse, then DN pulse, into 10 pF	Output voltage step, v_{gp} disturbance	$\approx \pm 10\text{ mV}$ ideal step
CP	Transient, lock condition	UP and DN together (150 ps), 10 pF	Net charge per cycle	≈ 0
CP	stb (replica loop)	V_c stepped over $0.2\text{--}0.9$ V	Loop phase margin	Stable at every V_c
Current reference	Parametric DC	R_S segment length	I_{REF}	$50\text{ }\mu\text{A}$
Current reference	DC op point, DC supply sweep	$V_{DD} = 1.0 / 1.1 / 1.2$ V	Start-up node, I_{REF} vs V_{DD}	Start-up off; low sensitivity
Current reference	Transient	V_{DD} ramp $0 \rightarrow 1.1$ V in $1\text{ }\mu\text{s}$	Start-up behaviour	Always starts
VCO	Transient + parametric	$V_c = 0.2\text{--}0.9$ V; W/L sweeps	f_{osc} vs V_c , K_{VCO}	$0.9\text{--}1.5\text{ GHz}$
VCO	PSS + pnoise	V_c at the lock point	Phase noise at 1 MHz , noise summary	$\leq -90\text{ dBc/Hz}$

Block	Experiment / analysis	Stimulus or swept variable	Measured quantity	Target
Divider	Transient	1.2 GHz input clock	Output and intermediate frequencies	75 MHz ($\div 16$)
Loop	AC on linear phase-domain model	–	Unity-gain frequency, phase margin	3.75 MHz, 60°
PLL_TOP	Closed-loop transient	Start from $V_c = 0$	f_{out} , lock time, V_c ripple, power	1.2 GHz, $\leq 8 \mu s$, $\leq 5 mW$
PLL_TOP	Transient + DFT	Locked window	Reference spur	$\leq -40 dBc$
PLL_TOP	Transient + edge extraction	Locked window	RMS period jitter	$\leq 8 ps$

Table 8: Simulation experiments performed on each block and on the complete PLL.

The analyses were chosen to match what each specification actually measures. The charge-pump currents are static quantities, so they were verified with DC sweeps, while the lock behaviour was checked only in closed-loop transient. The loop bandwidth and phase margin are small-signal properties of the locked loop. A locked PLL has no DC operating point that an AC analysis could linearise. PSS + PSTB was attempted on the full transistor-level PLL and did converge, but its loop gain showed a low-frequency slope of about $-15 dB/decade$ instead of the $-40 dB/decade$ expected of a type-II loop. That result was judged unreliable. The bandwidth and phase margin were therefore verified with an AC analysis of the linear phase-domain model (Figure 13), built from analogLib controlled sources, R and C, using the measured K_{VCO} and the PDK filter values.

4.3.2 Data Collection

Data were collected in Cadence Virtuoso ADE with the Spectre simulator. Each quantity was extracted as follows:

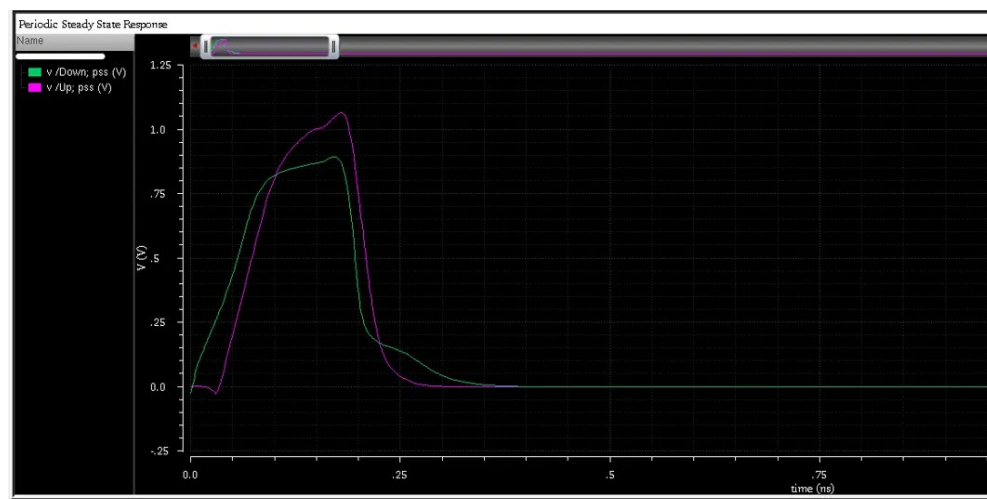
- **Charge-pump currents and mismatch.** From the DC sweep, I_{UP} and I_{DN} were read as the terminal currents of the output devices, and the mismatch $I_{UP} - I_{DN}$ was computed in the ADE calculator over 0.2–0.9 V. The values at the lock point ($V_c = 0.575 V$) were read with markers.
- **PFD transfer characteristic.** For each phase difference Δt , the charge-pump output current was integrated over one reference period to give the net charge per cycle, which was plotted against Δt . The slope was compared with the theoretical $I_{CP} = 0.05 fC/ps$.
- **Charge-pump steps and imbalance.** The output voltage steps were read from the transient waveform on the 10 pF load. The net charge per cycle was obtained as $Q = C \Delta V$.
- **Replica-loop stability.** An iprobe was inserted between the amplifier output (vgpa) and v_{gp} , and the stb analysis was run from 1 kHz to 10 GHz. The phase margin was read at each V_c , first at three points and then in 50 mV steps over 0.2–0.9 V.
- **VCO frequency and K_{VCO} .** The oscillation frequency was measured from the transient output after start-up had settled, with V_c swept parametrically. K_{VCO} was obtained as the derivative (deriv()) of the $f-V_c$ curve, read at the 1.2 GHz point and cross-checked with a finite difference between two markers.
- **Phase noise.** A PSS analysis of the free-running oscillator followed by pnoise gave the phase noise vs offset. The value was read at 1 MHz, and the flicker corner was read where the slope changes. The noise summary at 1 MHz, sorted by contributor, identified which devices dominate.

- **Loop bandwidth and phase margin.** From the AC analysis of the linear model, the unity-gain frequency was read where $|L(j\omega)| = 0$ dB and the phase margin as 180° plus the phase at that frequency.
- **Lock time and power.** Lock time was read from the closed-loop V_c transient as the time after which V_c settles to its final value, read with a marker (612.5 ns, Figure 37). The ADE settlingTime function with a 5 % tolerance (Figure 43) reports an earlier time, 520.9 ns. It only checks when V_c enters a ± 5 % band around its final value (about ± 29 mV). V_c reaches this band while it is still making small decaying oscillations, so the marker reading is the stricter measure and is the one quoted as the lock time. Power was calculated as VDD times the average supply current over the locked window, using the ADE expression $P = \text{abs}(\text{average}(\text{clip}(\text{IT}("/V0/PLUS"), 1.6 \mu\text{s}, 2 \mu\text{s}))) \times 1.1$.
- **Reference spur.** A DFT of the locked VCO output was taken over a 400 ns window. This is exactly 30 reference periods, so the DFT bins fall exactly on the carrier and on the $f_{out} \pm 75$ MHz sidebands. The spur level was read relative to the carrier.
- **RMS period jitter.** The RMS period jitter was calculated in the ADE calculator with the expression $\text{rms}(\text{period_jitter}(\text{clip}(v("/Vout"), 0.8 \mu\text{s}, 2 \mu\text{s}), \text{"rising"}, \text{threshold} = 0.55 \text{ V})))$. The function finds every rising crossing of the output through mid-supply (0.55 V) in the locked window, takes the differences between successive crossings as the periods, and returns the RMS deviation of these periods from their mean (Figure 39). The window starts at 0.8 μs , after lock, so the acquisition transient is excluded.

4.3.3 Results and Analysis

A. Phase-Frequency Detector

Figure 26 compares the UP and DN pulses at zero phase error before and after adding the reset delay cell and the output buffers. Without them, the pulses were triangular runt pulses about 150 ps wide, and DN reached only about 0.88 V. With them, both are full-swing 0–1.1 V pulses about 225 ps wide (Table 9).



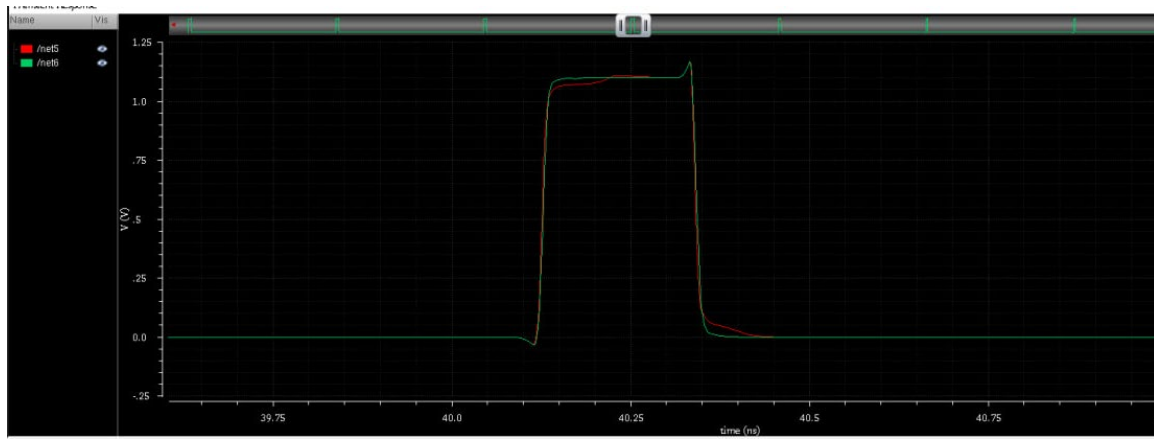


Figure 26: UP and DN pulses at zero phase error (a) before and (b) after adding the reset delay and output buffers.

Version	Pulse shape	Pulse width	Peak level
Without delay cell and buffers	Triangular (runt)	≈ 150 ps	DN ≈ 0.88 V
With delay cell ($L = 300$ nm) and buffers	Rectangular, full swing	≈ 225 ps	1.1 V (UP and DN)

Table 9: PFD output pulses at zero phase error.

Figure 27 shows the average charge delivered per reference cycle as a function of the phase difference Δt . The characteristic is a straight line with no flat region anywhere, including around $\Delta t = 0$. Its slope is 0.0525 fC/ps, against the theoretical 0.05 fC/ps for $I_{CP} = 50$ μ A (a 5 % difference). At zero phase error the curve sits at about $+6$ fC. This is a static phase offset caused by the UP > DN charge imbalance, not a dead zone. A separate test with a phase error of only 20 ps still produced a full UP pulse, widened by the reset delay (Figure 28).

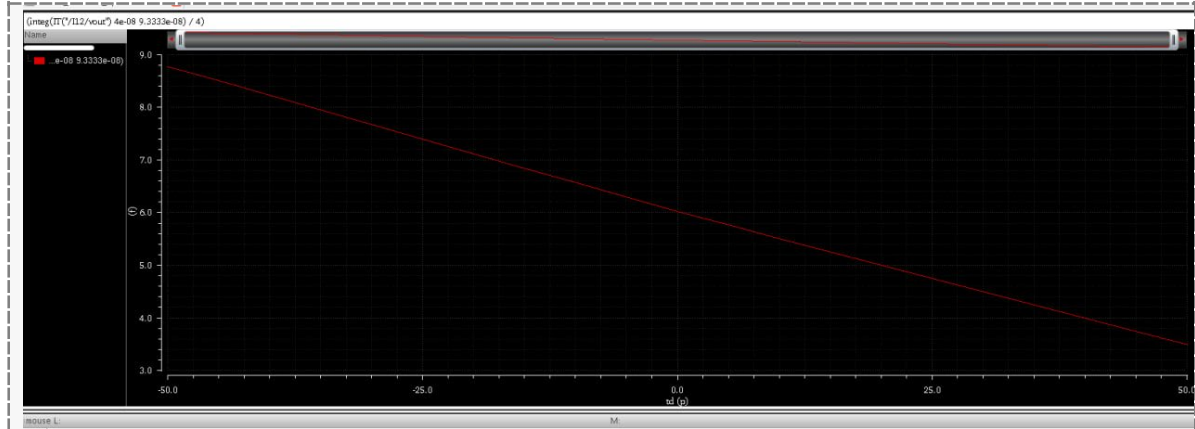


Figure 27: PFD + charge pump transfer characteristic: average charge per reference cycle vs phase difference.

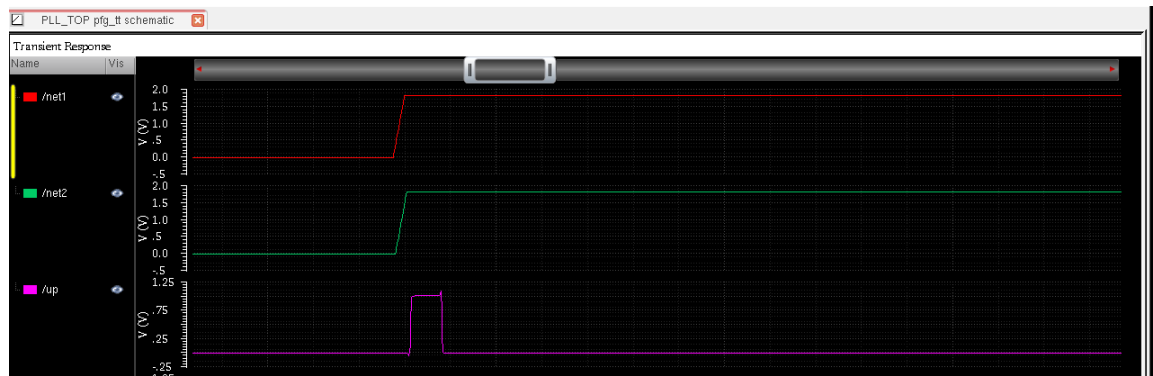


Figure 28: Dead-zone test with a 20 ps phase error between the inputs: a full-height UP pulse is still produced.

B. Charge Pump

All charge-pump test results are summarised in Table 10.

Test	Result
DC sweep, simple mirrors (min. size $\rightarrow$ L = 4 μ m sweep)	Best mismatch spread 8.1 μ A (allowed ± 2.5 μ A) $\times$
DC operating point, simple mirrors	Equal-V _{GS} devices carried 43.5 / 50.0 / 54.9 μ A, so channel-length modulation is the cause
DC sweep, replica bias (ideal reference)	Mismatch +1.07 / -0.83 μ A; I _{CP} = 50.0 μ A at lock $\checkmark$
Transient, separate pulses (L = 4 μ m sources)	$\approx \pm 45$ mV steps, ≈ 400 fC dumped per edge $\times$
Transient, separate pulses (L = 1 μ m sources)	+12.96 / -11.92 mV steps; v _{gp} moves 1.2 mV $\checkmark$
Reference: R _S length sweep	77 $\rightarrow$ 55.3 $\rightarrow$ 52.1 $\rightarrow$ 50.9 μ A; final segL 5.235 μ m $\rightarrow$ 51.1 μ A $\checkmark$
Reference: DC operating point	Start-up node 43.6 mV (start-up device off) $\checkmark$
Reference: supply sweep (1.0 / 1.1 / 1.2 V)	47.3 / 51.1 / 54.8 μ A ($\approx \pm 7$ %)
Reference: start-up transient	Turns on at VDD ≈ 0.6 V, settles at ≈ 51 μ A $\checkmark$
DC sweep, integrated CP with on-chip reference	Mismatch +0.892 / -0.519 μ A; I _{UP} = 50.02 μ A, I _{DN} = 50.00 μ A at lock $\checkmark$
stb, replica loop (before compensation)	PM 66.8° / 11.9° / 75.4° at V _c = 0.2 / 0.575 / 0.9 V $\times$
stb, replica loop (after compensation)	PM $\geq 68^\circ$ over 0.2–0.9 V (minimum at 0.55 V), 74° at lock $\checkmark$
DC sweep after compensation	Mismatch +0.77 / -1.27 μ A (at 0.2 / 0.9 V), 50.0 μ A at lock $\checkmark$
Transient, lock condition (both pulses 150 ps)	≈ 14 fC/cycle residual imbalance (UP > DN)

Table 10: Summary of charge pump tests and results.

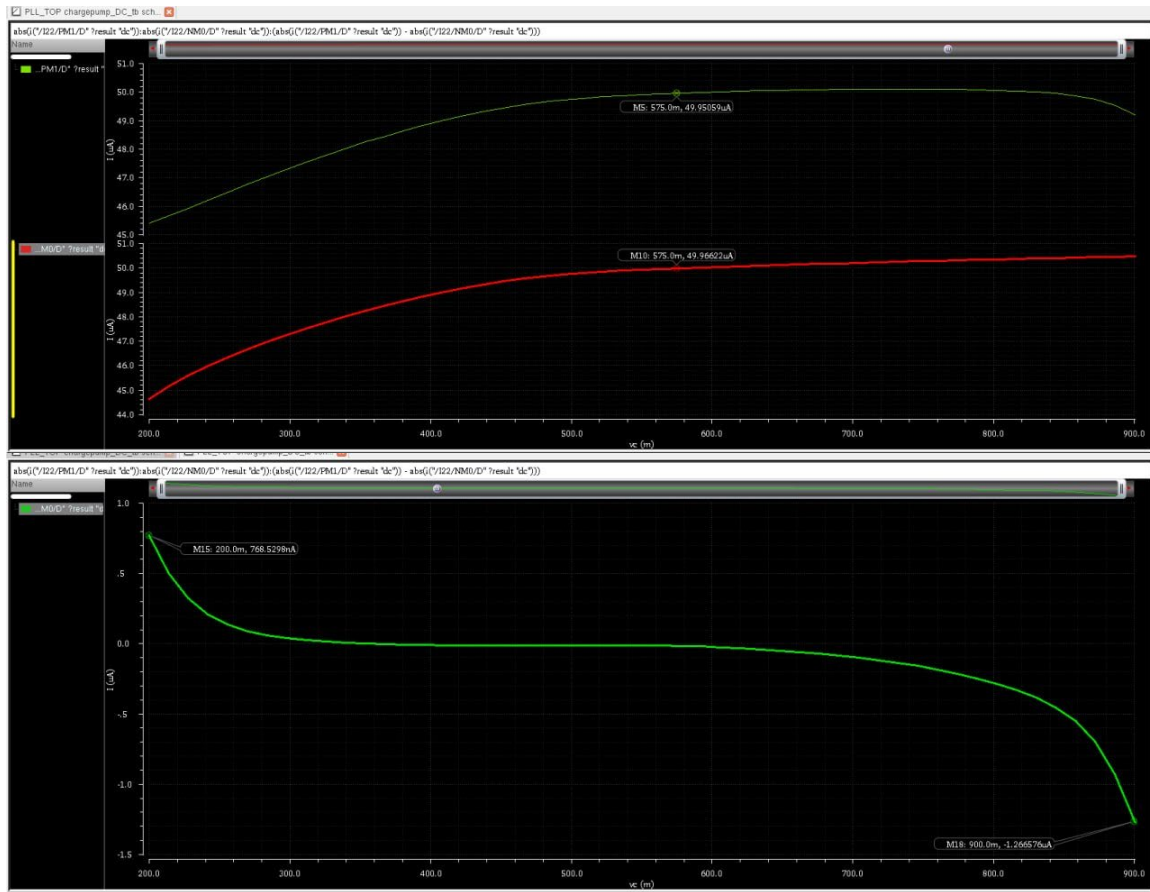


Figure 29: Replica-bias charge pump: (a, top) UP (green) and DN (red) currents and (b, bottom) UP – DN current mismatch vs output voltage.

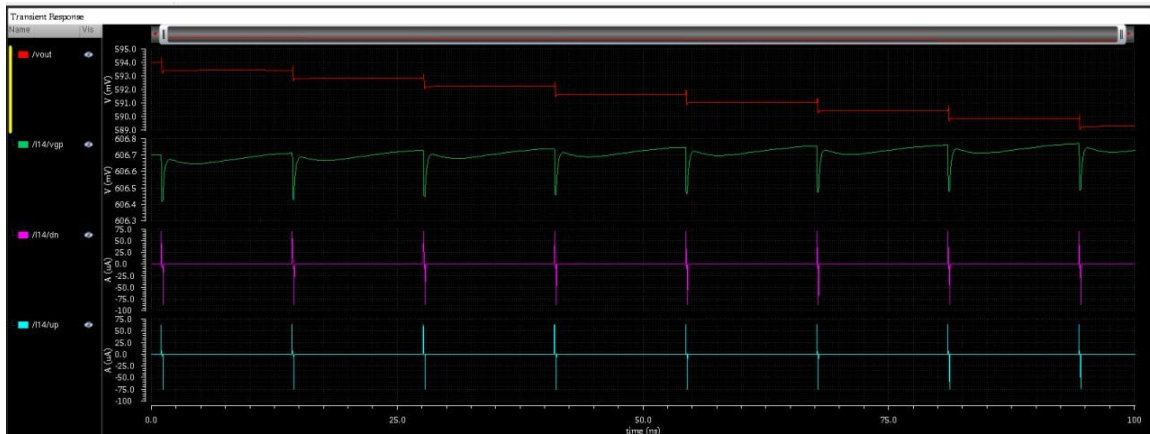


Figure 30: Charge pump transient response (final design): output voltage, v_{gp} and the UP and DN branch currents.

The compensation of the replica loop is shown step by step in Table 11 and Figure 31.

Change	PM at 0.575 V
Initial design	11.9°
OTA tail currents $1 \mu\text{A} \rightarrow 0.25 \mu\text{A}$	44.1°
OTA mirror loads $4\times$ narrower	51.6°
MOS capacitor on v_{gp} : 4 / 8 / 12 / 16 fingers	51.6° / 65° / 71.5° / 75.4°
Final: 14 fingers (fine sweep)	74° at lock; worst case 68° at 0.55 V

Table 11: Replica-loop phase margin at the lock point ($V_c = 0.575 \text{ V}$) after each compensation step.

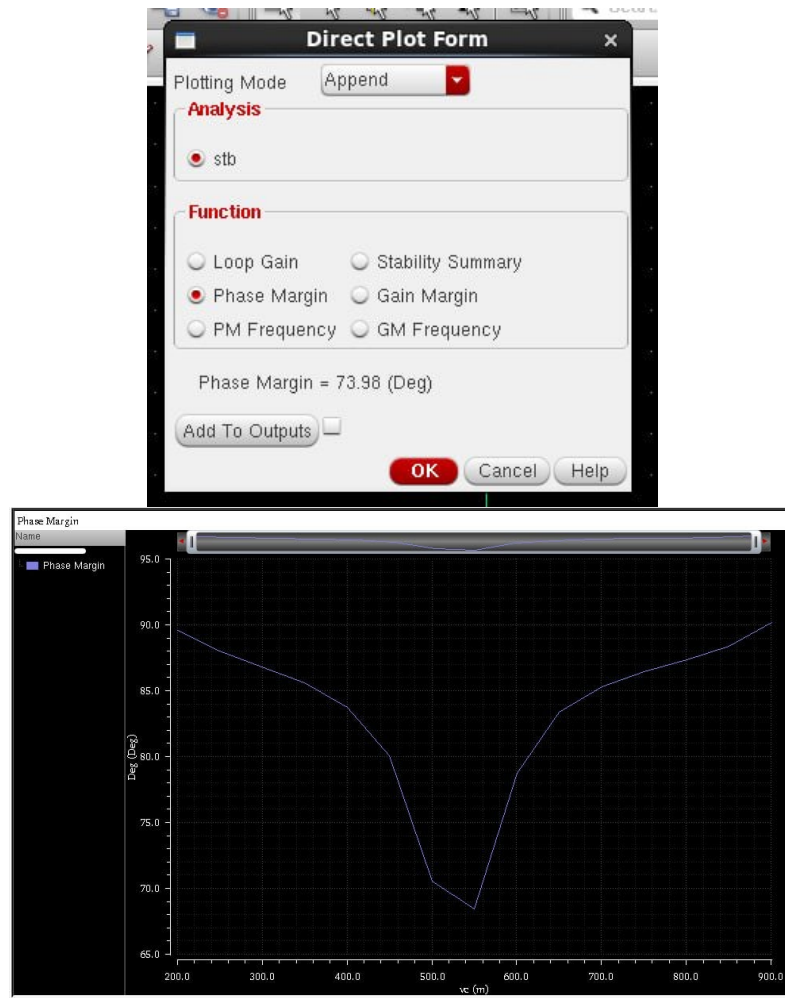


Figure 31: Replica-loop stability (final design): (a, top) stb phase margin at the lock point and (b, bottom) phase margin vs V_c over 0.2–0.9 V.

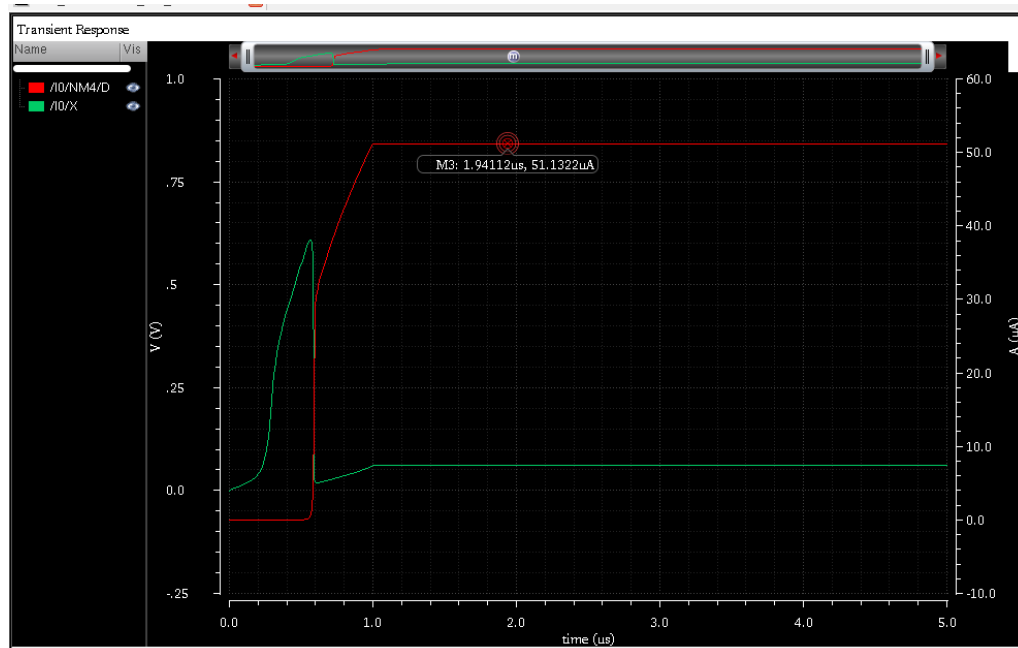


Figure 32: Start-up transient of the beta-multiplier reference; the reference current settles at 51.13 μA .

C. Voltage-Controlled Oscillator

The measured tuning curve (Figure 33) covers 900.7 MHz to 1.50 GHz over $V_c = 0.2\text{--}0.9$ V. This covers the required 0.9–1.5 GHz. The VCO reaches 1.2 GHz at $V_c \approx 0.58$ V (575.7 mV in the locked PLL), where the local slope is $K_{VCO} \approx 1.08$ GHz/V. The phase-noise improvement steps are listed in Table 12, and the final phase-noise plot is shown in Figure 34.

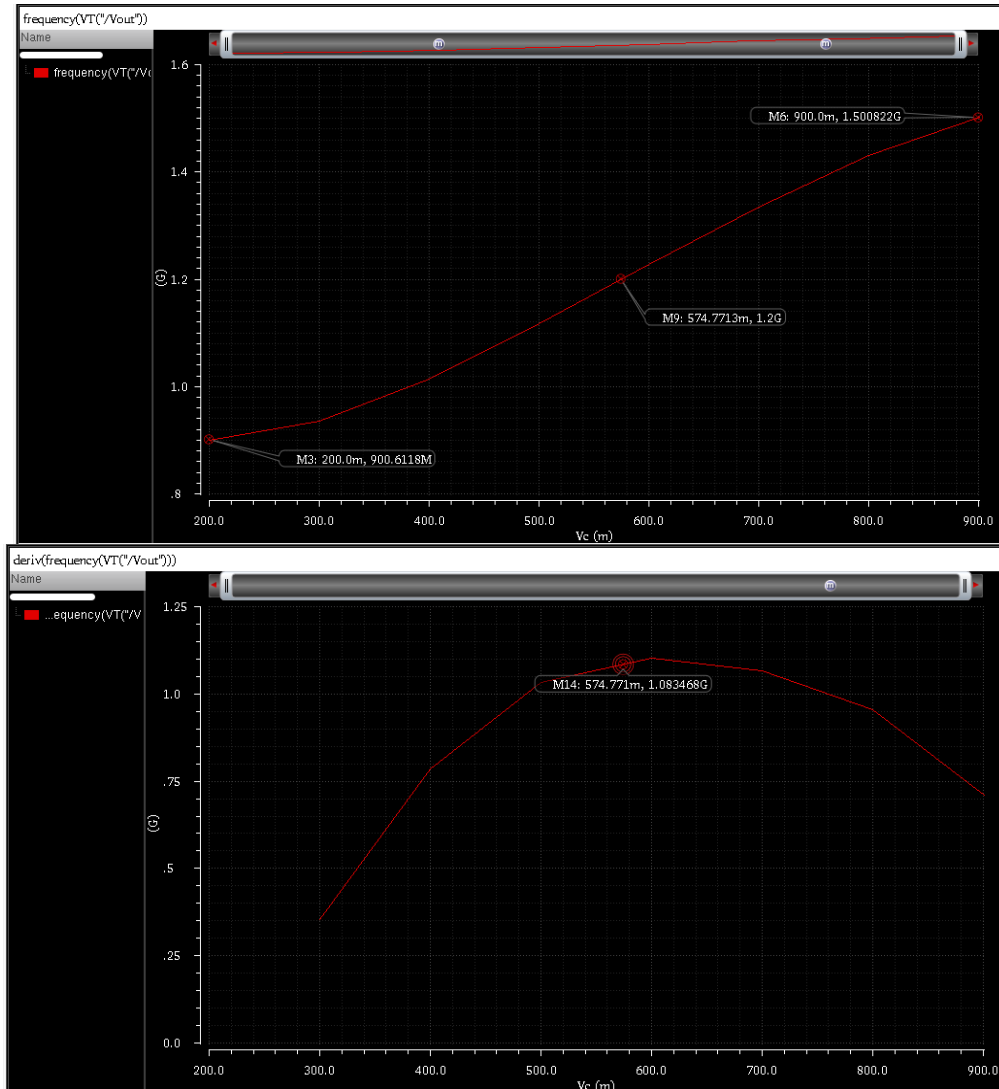


Figure 33: (a, top) VCO output frequency vs V_c over 0.2–0.9 V and (b, bottom) frequency and its derivative K_{VCO} around the lock point.

Design step	Phase noise at 1 MHz	Flicker corner	VCO power
Initial design	−76.8 dBc/Hz	> 1 MHz	≈ 150 μW
Bias/mirror L: 45 nm → 1 μm	−82 dBc/Hz	255 kHz	≈ 150 μW
All W × 12, all R ÷ 12	−92.8 dBc/Hz	45.7 kHz	≈ 1.8 mW
Final VCO as integrated in the PLL	−91.9 dBc/Hz	—	—

Table 12: VCO phase noise at 1 MHz offset after each design step.

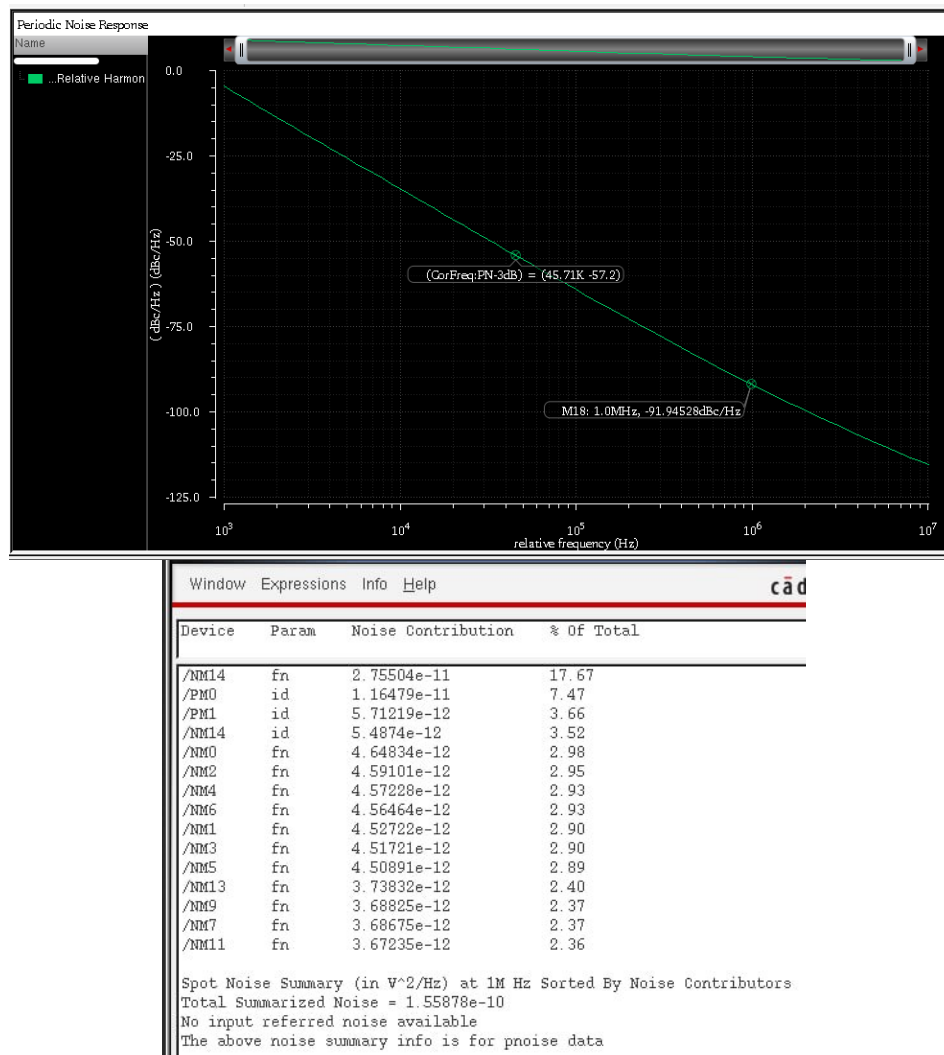


Figure 34: (a, top) Final VCO phase noise vs offset frequency (-91.95 dBc/Hz at 1 MHz) and (b, bottom) noise summary at 1 MHz sorted by contributor.

D. Frequency Divider

Figure 35 shows the divider driven by a 1.2 GHz clock. Each stage halves the frequency, and the output is 75 MHz. In the closed-loop simulation the PLL locks with the VCO at 1.2000 GHz against the 75 MHz reference, which confirms a division ratio of exactly 16.

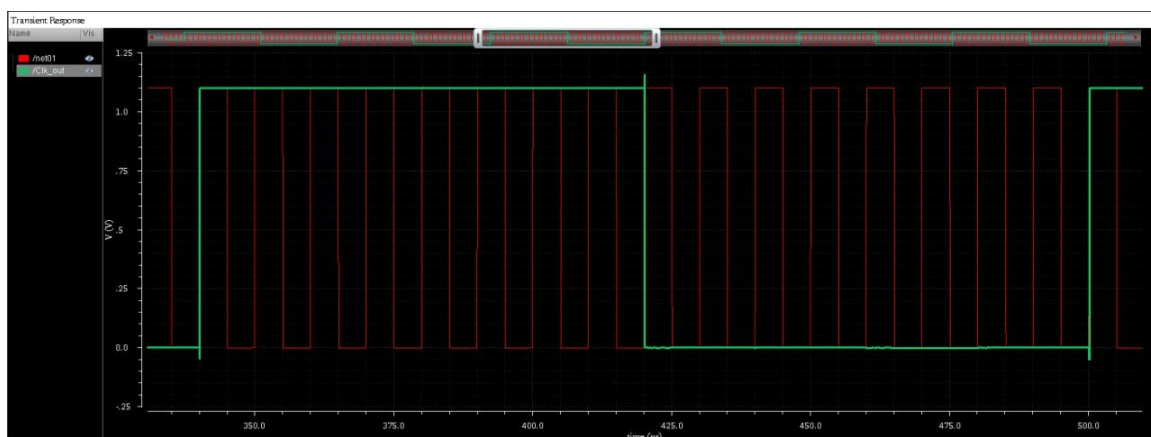


Figure 35: Divide-by-16 transient: 16 cycles of the input (red) equal one cycle of the output (green).

E. Loop Dynamics

The AC analysis of the linear phase-domain model (Figure 13), using the measured K_{VCO} and the PDK loop-filter values, gives a unity-gain frequency of 3.752 MHz and a phase margin of 60.01° (Figure 36).

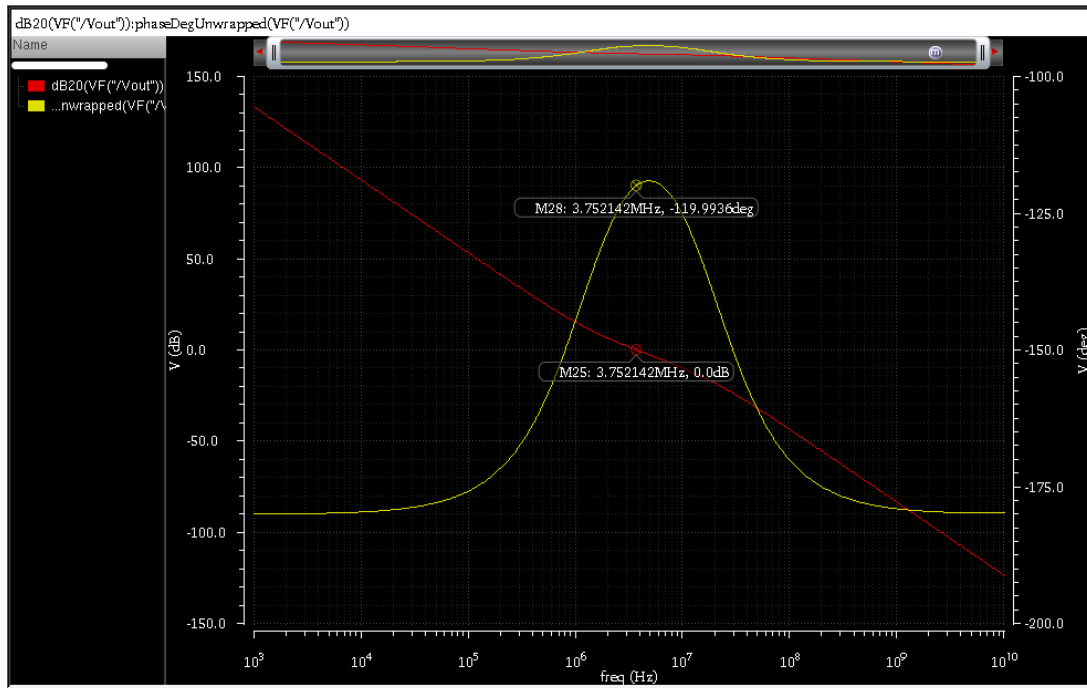


Figure 36: Open-loop gain and phase of the linear phase-domain model with PDK loop-filter values.

F. Complete PLL

Figure 37 shows the closed-loop acquisition. V_c starts from zero, rises as the loop pulls the VCO up, and settles at 575.7 mV, with lock reached at 612.5 ns. In lock, V_c shows a residual ripple of 10.13 mV peak-to-peak at the reference frequency. Several integration problems were found and corrected during the full-PLL simulations (Table 13).

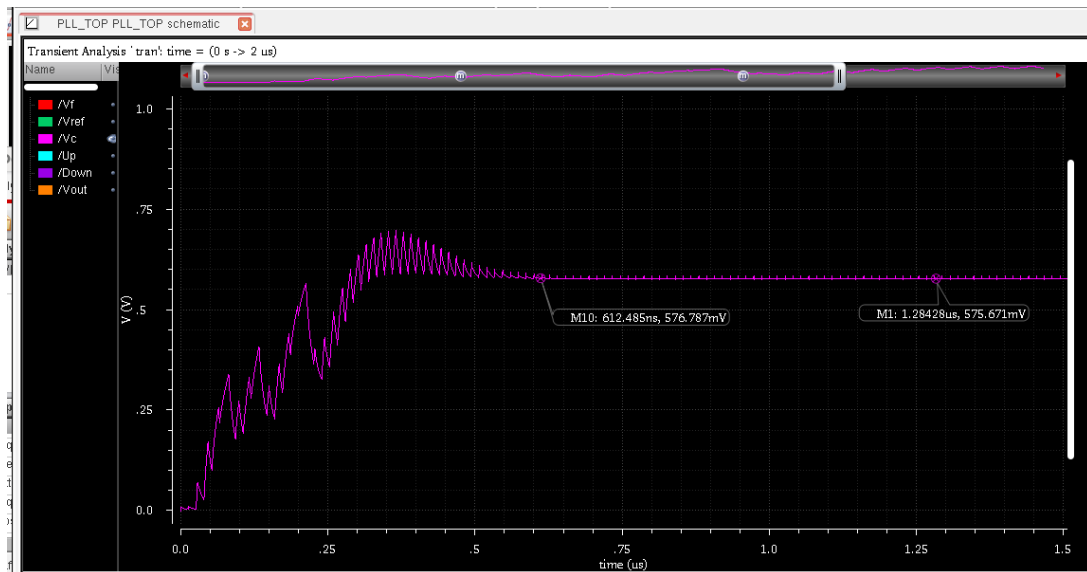


Figure 37: Closed-loop acquisition: V_c vs time. The loop locks in 612.5 ns and V_c settles at 575.7 mV.

Observation	Cause	Fix
V _c pinned at VDD, no lock	CP polarity: PMOS switch conducting continuously	Switch drive polarity corrected
Divider not working	VDD and GND swapped on the divider	Supply connections corrected
Reference source period wrong	vpulse period expression syntax	Unit suffix placed on the numerator
V _c ripple ≈ 21 mV in lock	Charge dumped from UP source PM0 entering triode	PM0/PMR resized $20 \times 1.25 \mu\text{m} / 1 \mu\text{m} \rightarrow 10 \times 1.25 \mu\text{m} / 0.5 \mu\text{m}$; ripple 3.7 mV; replica PM re-checked ($\geq 68^\circ$ over 0.2–0.9 V)
Ripple rose to ≈ 10 mV after the PFD reset delay change	Longer overlap of UP and DN with residual mismatch; imbalance reversed (UP wider than DN)	Final ripple 10.13 mV p-p; reference spur -69.4 dBc, within the -40 dBc limit

Table 13: Problems found during full-PLL integration and their fixes.

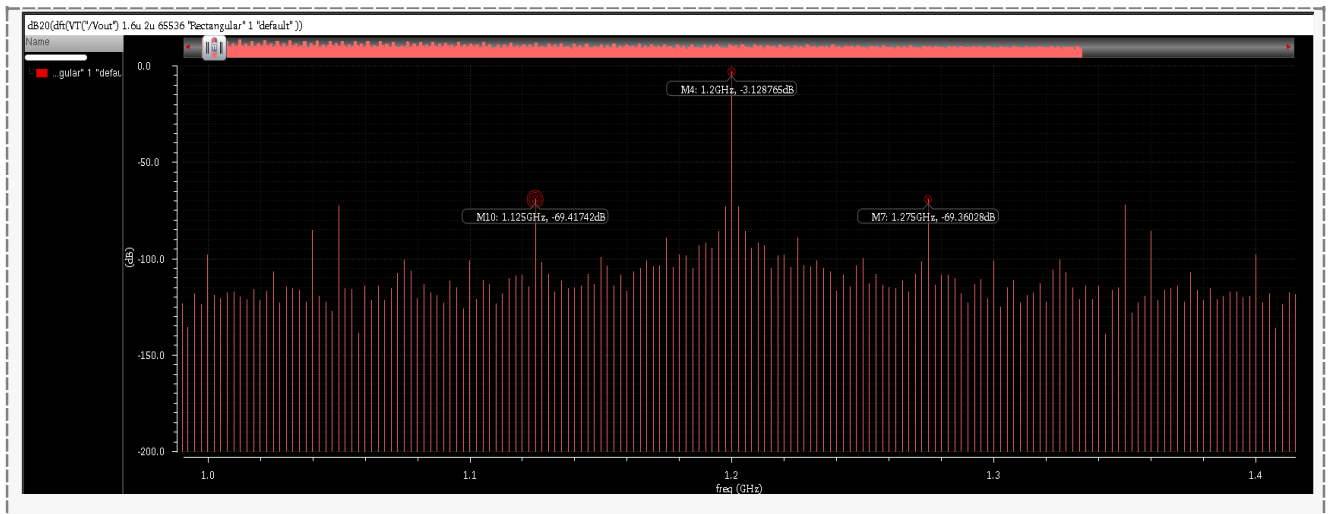


Figure 38: Spectrum of the locked output showing the reference spur level.

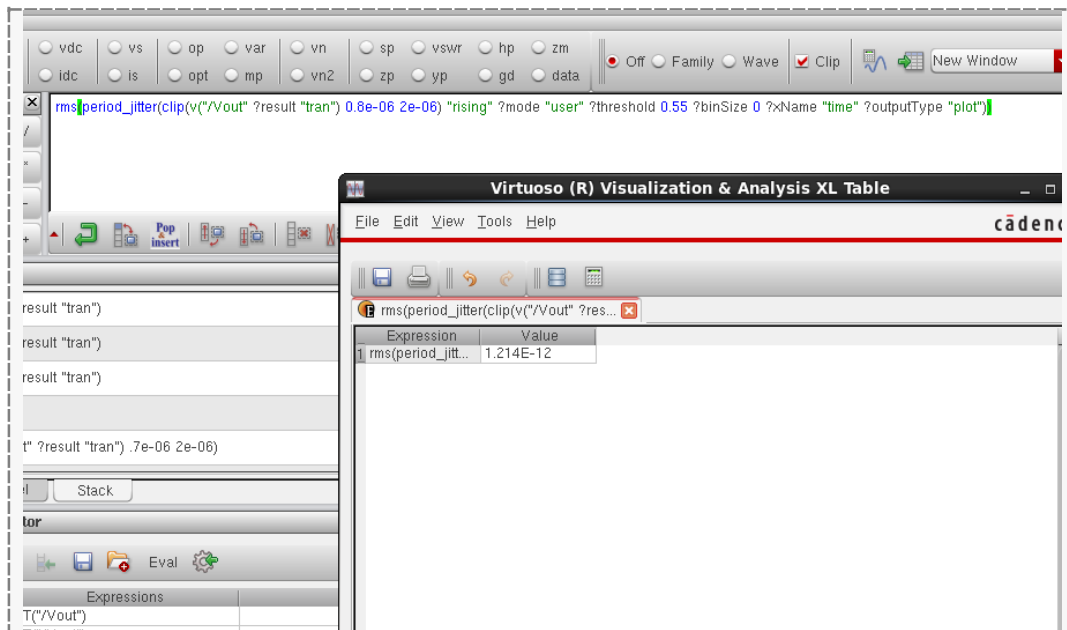


Figure 39: RMS period jitter calculated in the ADE calculator with `rms(period_jitter(...))` over the locked window (0.8–2 μs), giving 1.214 ps.

After lock, the divider output and the 75 MHz reference have the same frequency (Figure 40), and the

VCO output is 1.200 GHz (Figure 41). UP and DN become narrow pulses once per reference period (Figure 42). The lock time, V_c at lock, ripple and power were read from the ADE outputs (Figure 43), and the complete layout passes LVS (Figure 44).

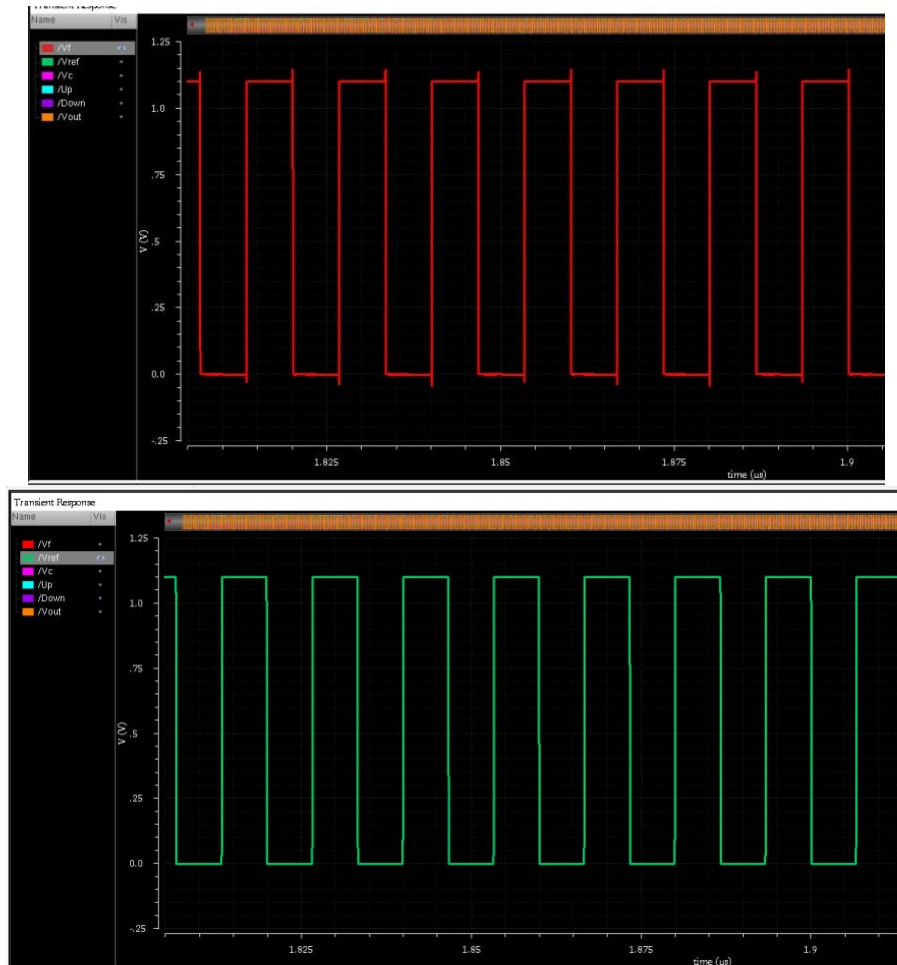


Figure 40: After lock: (a, top) divider output (feedback) and (b, bottom) 75 MHz reference, at the same frequency.

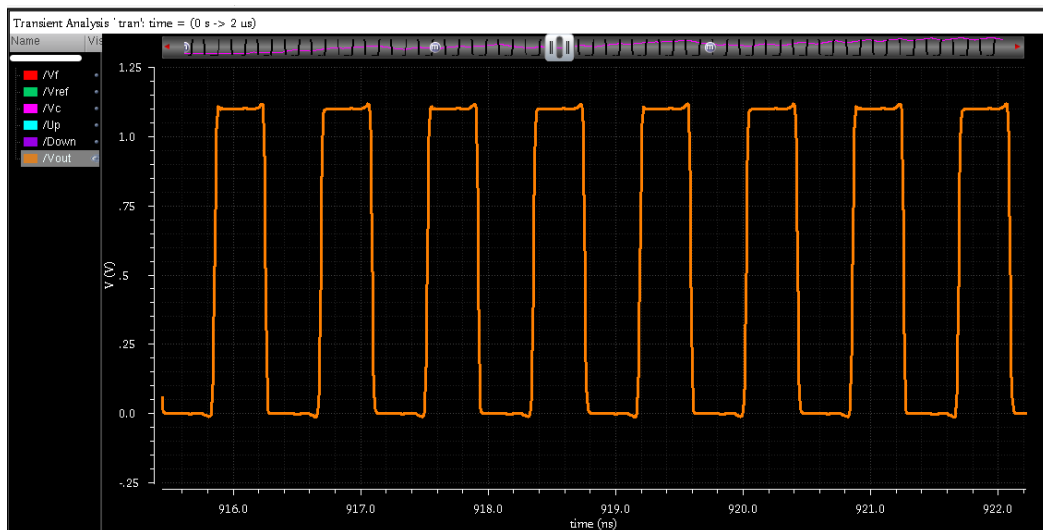


Figure 41: VCO output after lock (1.200 GHz).

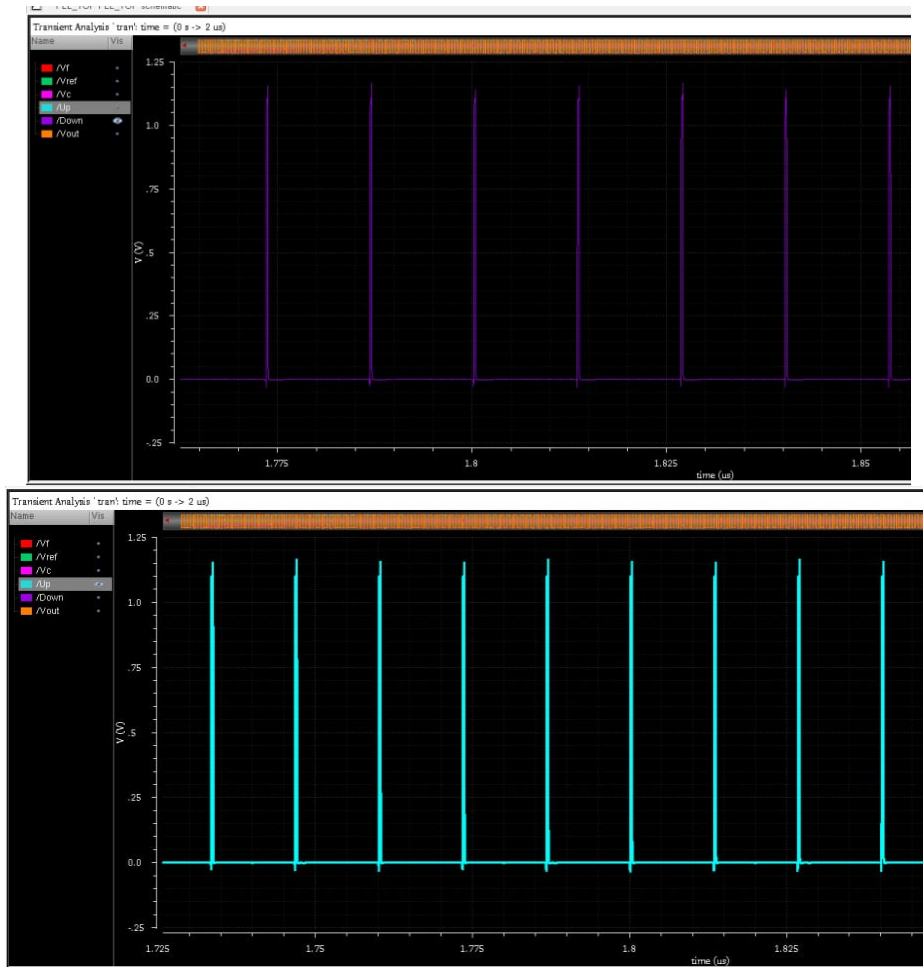


Figure 42: Charge-pump control pulses at lock: (a, top) DN and (b, bottom) UP.

Outputs					
	Name/Signal/Expr	Value	Plot	Save	Save Options
1	Vf		☒	☐	allv
2	Vref		☒	☐	allv
3	Vc		☒	☐	allv
4	Up		☒	☐	allv
5	Down		☒	☐	allv
6	Vout		☒	☐	allv
7	frequency(clip(VT("/Vf") 1.6e-06 2e-06))	75M	☒	☒	
8	frequency(clip(VT("/Vout") 1.6e-06 2e-06))	1.2G	☒	☐	
9	average(clip(VT("/Vc") 1.6e-06 2e-06))	575.7m	☒	☐	
10	settlingTime(VT("/Vc") 0 t 2e-06 t 5)	520.9n	☒	☐	
11	frequency(clip(VT("/Vf") 8e-07 1.2e-06))	75M	☒	☐	
12	V0/PLUS		☐	☒	yes
13	(abs(average(clip(IT("/V0/PLUS") 1.6e-06 2e-06))) * 1.1)	2.315m	☒	☐	
14	(ymax(clip(VT("/Vc") 1.6e-06 2e-06)) - ymin(clip(VT("/Vc") 1.6e-06 2e-06)))	10.13m	☒	☐	

Figure 43: ADE outputs after lock: reference and output frequency, V_c at lock (575.7 mV), 5 % settling time (520.9 ns; the lock time of 612.5 ns is explained in Section 4.3.2), power (2.315 mW) and V_c ripple (10.13 mV p-p).

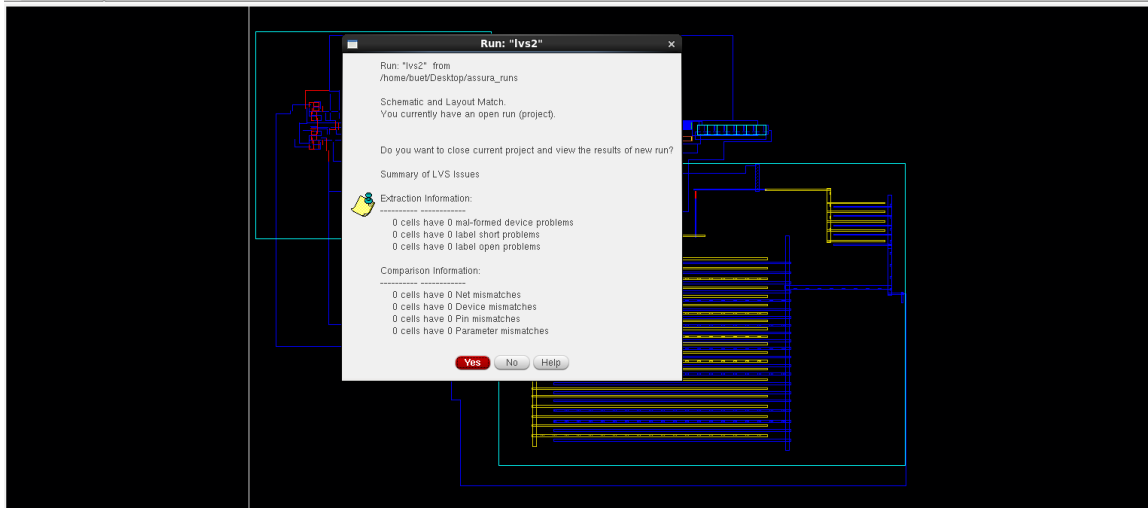


Figure 44: LVS result for the complete PLL layout: schematic and layout match.

The locked output frequency is 1.200 GHz (16×75 MHz). The reference spur at $f_{\text{out}} \pm 75$ MHz is -69.4 dBc, the RMS period jitter is 1.214 ps, and the total power after lock is 2.315 mW. The complete layout is DRC and LVS clean, with an area of $933.09 \mu\text{m} \times 669.89 \mu\text{m} = 0.6251 \text{ mm}^2$. The final simulated performance is compared with the specifications in Table 14.

Parameter	Specification	Simulated	Status
Output frequency	1.20 GHz	1.200 GHz	✓
VCO tuning range	0.9–1.5 GHz	900.7 MHz – 1.50 GHz	✓
K _{VCO} (at lock)	750 MHz/V	≈ 1.08 GHz/V (loop filter rescaled)	Compensated
Charge pump current	50 μA	50.0 μA at lock	✓
CP mismatch (0.2–0.9 V)	$\leq 5\%$ (2.5 μA)	$+0.77 / -1.27 \mu\text{A}$ ($\leq 2.5\%$)	✓
PFD dead zone	None	None (20 ps test; linear charge vs Δt)	✓
Loop bandwidth	3.75 MHz	3.752 MHz	✓
Phase margin	60°	60.01°	✓
Lock time	$\leq 8 \mu\text{s}$	0.61 μs (612.5 ns)	✓
VCO phase noise at 1 MHz	≤ -90 dBc/Hz	-91.9 dBc/Hz	✓
Reference spur	≤ -40 dBc	-69.4 dBc	✓
RMS period jitter	≤ 8 ps	1.214 ps	✓
Total power	≤ 5 mW	2.315 mW	✓
Layout	DRC and LVS clean	DRC and LVS clean; $933.09 \times 669.89 \mu\text{m} = 0.6251 \text{ mm}^2$	✓

Table 14: Simulated performance against specifications.

4.3.4 Interpretation and Conclusions on Data

PFD. A dead zone appears as a flat region of the transfer characteristic around zero phase error, where small errors produce no charge. The measured characteristic is a straight line with no flat region anywhere, and its slope (0.0525 fC/ps) agrees with the theoretical $I_{CP} = 0.05$ fC/ps to within 5 %. The PFD is therefore free of a dead zone. This confirms that the reset delay ($L = 300$ nm inverters) makes the minimum pulse longer than the switch turn-on time, and that the output buffers deliver full-swing pulses to the charge pump. The curve does not pass through the origin: at zero phase error it sits at about +6 fC. This is a static phase offset caused by the $UP > DN$ charge imbalance in the charge pump,

not a dead zone, because the slope stays constant through $\Delta t = 0$.

Charge pump. The simple-mirror results show that the mismatch is a structural problem (channel-length modulation under unequal V_{DS}) that device sizing cannot solve within a 1.1 V supply. The replica-bias architecture solves it by equalising the operating conditions: the worst-case mismatch of $+0.77 / -1.27 \mu\text{A}$ is at most 2.5 %, half of the 5 % limit, and the current at lock is exactly $50.0 \mu\text{A}$. The stb results show that the extra feedback loop must be checked at every operating point. The original design was stable at both ends of the range but had only 11.9° of phase margin at the lock point, where both OTAs are active. After compensation, the minimum margin is 68° (at 0.55 V) and the margin at the lock point is 74° . The beta-multiplier reference delivers $51.1 \mu\text{A}$ with about $\pm 7\%$ supply sensitivity and starts reliably. Because the replica loop sets the matching, this small error in the absolute current only scales I_{CP} slightly and does not affect UP/DN matching. The residual imbalance appears in the full PLL as a ripple of 10.13 mV peak-to-peak on V_c . It is the price of the longer reset pulse that removes the dead zone, since UP and DN now overlap for about 225 ps every cycle.

VCO. The phase-noise target is met with a 1.9 dB margin (-91.9 dBc/Hz). The tuning range covers the required 0.9–1.5 GHz. The noise results confirm the two-mechanism analysis. Lengthening only the bias devices improved the noise by about 5 dB, the expected limit once the flicker corner falls below the 1 MHz spec point. Impedance scaling by 12 gave a further 10.8 dB, matching the predicted $10\log_{10} 12$, at the cost of about 12 times the VCO power. The measured K_{VCO} ($\approx 1.08 \text{ GHz/V}$) is higher than the 750 MHz/V target. As shown in Section 3.2, this is unavoidable: covering 0.9–1.5 GHz within 0.2–0.9 V needs at least 857 MHz/V on average. The deviation was compensated at system level by rescaling the loop filter, and the loop specifications are met.

Loop and complete PLL. The linear model gives 3.752 MHz and 60.01° , matching the 3.75 MHz / 60° design. The measured lock time of $0.61 \mu\text{s}$ is close to the $\approx 0.5 \mu\text{s}$ minimum predicted from the slew-rate limit plus the linear settling time. Figure 37 shows that V_c rises in a sawtooth during acquisition. Several times between about 0.05 and $0.3 \mu\text{s}$ it falls back briefly. These are cycle slips: the VCO phase moves a full reference cycle ahead of or behind the reference while the frequency is still wrong, so the PFD output briefly reverses. V_c then overshoots to about 0.7 V before settling. The cycle slips and the overshoot add only about $0.1 \mu\text{s}$ to the minimum estimate, so the loop behaves as designed. All system-level specifications are met, most with large margins: lock time is about 13 times faster than required, jitter 6.6 times lower, power at 46 % of the budget, and the reference spur 29.4 dB below the limit.

Limitations of the verification. All results are for the typical process corner at nominal supply. The linear model is valid only for small phase errors near lock and does not capture acquisition behaviour, which was checked separately in transient. PSS + PSTB on the full transistor-level PLL did not give a credible loop response. Corner and temperature simulations, and post-layout simulation with extracted parasitics, are needed before the design can be considered robust.

4.4 Limitations of Tools (PO(e))

We used Cadence Virtuoso, Spectre, the GPDK045 process design kit and layout verification tools to design and check the PLL. These tools helped us find circuit problems before fabrication. However, the reported results are predictions from models, so they do not prove the performance of a manufactured chip.

Operating conditions. The results in this project mainly use the typical process corner and a 1.1 V supply. The on-chip current reference was also checked at 1.0 V and 1.2 V, but this is not a complete test of the PLL under supply variation. Process corners, temperature changes and random device mismatch still need separate tests.

Model limits. The reported PDK model limits [4] include a minimum channel length of 40 nm and a maximum width of 10 micrometres per finger. Wider transistors must use several fingers. These are device model and layout limits, not a guarantee that every possible size will give accurate or suitable circuit behaviour.

Time steps and tolerances. A simulator calculates voltages at selected time points. If the time step is too large, it can miss short pulses or give inaccurate edge times. This matters here because the PFD pulses are about 225 ps wide and the output period is about 833 ps. Solver tolerances and waveform saving settings can also affect calculated jitter, lock time and ripple. The exact numerical error is not established by the present tests.

Limits of the analysis methods. Direct periodic steady-state and periodic stability analysis of the full PLL did not give a reliable loop response. We therefore used a separate linear phase model to check bandwidth and phase margin, and transient simulation to check start-up and lock. The linear model applies near lock; it cannot describe large errors or cycle slipping during start-up.

Measurement uncertainty. Noise and spectrum results depend on the analysis settings and the time window used after lock. The reported 1.214 ps RMS period jitter applies to the selected locked window. It does not include every effect that a real reference clock, supply or package could introduce. The ideal reference source also leaves out reference clock noise.

Physical verification. The layout passed design rule checking (DRC) and layout versus schematic checking (LVS). DRC checks whether the geometry follows the rules. LVS checks whether the layout connections match the schematic. Neither check proves that the layout has the same speed, noise or lock time as the schematic. Post-layout simulation with extracted resistance and capacitance is still needed.

Component tolerances. The filter uses PDK resistor and capacitor devices. Their available values are not perfectly continuous, and the resistor includes end and contact resistance. Manufacturing variation can change these values further. We adjusted the filter using the measured VCO gain, but the complete tolerance range has not yet been verified.

4.5 Impact Assessment (PO(f))

4.5.1 Assessment of Societal and Cultural Issues

A PLL creates a fast clock from a slower reference. Such clocks can support communication equipment, processors and other electronic systems. Our circuit produces a 1.20 GHz clock from a 75 MHz reference. The possible benefit is a compact timing block that could be included in a larger chip after further testing.

The project also gives practical experience in analog circuits, digital circuits and IC layout. This knowledge can help students develop useful electronic systems. A commercial benefit cannot be claimed from simulation alone. It would also depend on reliability, manufacturing cost and the needs of the final users.

The PLL itself does not collect personal information or depend on a user's language, religion or cultural background. When it is used in a larger product, that product should have clear instructions and should be usable by its intended community. Simple explanations are useful when presenting the project to people without a circuit design background.

4.5.2 Assessment of Health and Safety Issues

The present project is a computer-based design and layout exercise. It does not involve a manufactured chip or direct connection to mains power. The simulated circuit runs at 1.1 V and uses 2.315 mW after lock. These figures describe the circuit model; they do not describe the electrical safety of the computer or laboratory equipment.

If a chip is built later, it should be tested with a regulated, current-limited supply. The supply polarity and pin connections must be checked before power is applied. Protection against electrostatic discharge, short circuits and excessive temperature will also be needed. A clock used in a medical, transport or other safety-related product would require much wider reliability testing.

4.5.3 Assessment of Legal Issues

The software, process files and design references should be used under their applicable licences and institutional rules. Permission to use a tool for coursework should not be treated as permission to redistribute its installation files or process models. A public project folder should contain only material that the team is allowed to share.

The report should acknowledge published circuit ideas and reference material. Before commercial use, the design would also need a review of relevant intellectual property and the requirements of the selected manufacturing process. GPDK045 is a generic design kit, so its simulated results should not be presented as a fabrication-ready commercial qualification.

4.6 Sustainability Evaluation (PO(g))

The final PLL uses 2.315 mW in the reported locked simulation, which is below the 5 mW design limit. At constant operation, this circuit-level power corresponds to about 0.0556 Wh in 24 hours. This calculation excludes the reference source, supporting circuits and the computer used for simulation.

Low circuit power can reduce energy use when many such clock blocks operate in electronic products. There is a clear trade-off between power and clock quality. The VCO design increased transistor widths and bias current to improve phase noise. This used more power, but the final PLL still stayed within the specified budget. A further reduction in power should be accepted only if the required noise, tuning range and lock behaviour are maintained.

Simulation allows design faults to be corrected before a chip is made. This can avoid unnecessary physical prototypes. However, simulations still use computer energy, and long runs should have a clear purpose. Semiconductor fabrication, packaging and disposal would add material use and waste if the design moves to hardware.

The reported core layout area is 0.6251 square millimetres. A smaller area could reduce material use and manufacturing cost, but it must not weaken matching or reliability. A full sustainability assessment would need manufacturing, packaging and lifetime data. At this stage, the strongest measurable environmental result is the low simulated operating power.

4.7 Ethical Issues (PO(h))

The main ethical responsibility in this project is to describe the design honestly. We distinguish calculated values, schematic simulation results and layout checks. We do not describe simulated waveforms as measurements from a fabricated chip. This lets the reader judge what has actually been verified.

One reporting challenge was the VCO gain. Its value near lock was about 1.08 GHz/V, above the 750 MHz/V target. We kept this difference visible and adjusted the loop filter for the measured gain. The resulting bandwidth was 3.752 MHz and the phase margin was 60.01 degrees. Calling the original gain target fully met would hide an important design trade-off.

The complete PLL results apply to the stated simulation conditions. They should not be extended to all temperatures, supplies or process corners without evidence.

The early charge pump had poor current matching, and the first replica loop had a low phase margin near the lock point. These results were useful evidence. The design was changed to improve matching and stability rather than presenting only the successful final plots. The report records the main problems and their fixes.

Circuit ideas taken from textbooks, tutorials and process manuals should receive proper credit. Each member's contribution must also be recorded accurately. Names should not be attached to work that a person did not perform. The academic honesty statement in the main report must be completed and signed according to the course instructions.

Results should remain traceable to the final circuit, test setup and analysis settings. Keeping the final design files and waveforms helps another person check the work. Any shared files must also respect software and PDK permissions. These practices support honest reporting, fair credit and reproducible engineering work.

5 Reflection on Individual and Team work (PO(i))

The project required both separate block work and complete-system checks. A working PFD, charge pump, VCO or divider alone was not enough. Their connections, signal levels and gains had to agree before the PLL could lock. This showed us why individual responsibility and shared review are both important.

5.1 Individual Contribution of Each Member

The main work areas were circuit calculations, transistor-level design, block testing, full-loop integration, layout, DRC and LVS, and report preparation. A contribution record should connect each member to the work and evidence that they personally produced. Table 15 gives this record for each member.

Member Student ID	Actual contribution	Evidence of work
2106089	Worked on the replica-bias charge pump and bias circuits, seven-stage current-starved VCO, and four-stage divide-by-two divider. Helped prepare their layouts, the report and the presentation.	Charge-pump error amplifier and beta-multiplier reference; current matching and stability (STB) sweeps; VCO tuning range, gain and phase-noise checks; divider functional verification.
2106150	Worked on the tri-state PFD, charge pump and bias circuits, and seven-stage VCO. Helped prepare their layouts and integrate the complete PLL. Contributed to the report and presentation.	PFD reset-delay cell, output buffers and dead-zone tests; replica-bias charge pump, error amplifier and current reference; matching and stability sweeps; VCO tests; lock, jitter, spur and power results; DRC and LVS checks.
2106181	Worked on the charge pump and bias circuits, loop filter, and four-stage divide-by-two divider. Helped prepare their layouts, the report and the presentation.	Replica-bias charge pump, error amplifier and beta-multiplier reference; matching and stability sweeps; loop-filter calculation and rescaling for VCO gain; mimcap and poly-resistor implementation; divider functional verification.
2106182	Worked on the tri-state PFD and loop filter. Helped prepare their layouts and assemble the complete PLL. Contributed to the report and presentation.	PFD reset-delay cell, output buffers and dead-zone tests; loop-filter calculation and rescaling for VCO gain; mimcap and poly-resistor implementation; lock, jitter, spur and power results; DRC and LVS checks.

Table 15: Individual contribution record.

Block design needs careful device sizing and local tests. Integration needs a clear understanding of the whole loop. Layout work needs patient checking of connections and process rules. Reporting needs correct units and a clear explanation of the results. These responsibilities are different, but each affects the quality of the final project.

5.2 Mode of Teamwork

The technical work followed a block-by-block approach. Each block was checked separately before the complete PLL was tested. Changes to one block sometimes required changes elsewhere. For example, the measured VCO gain changed the loop filter values, and the PFD reset delay affected control-voltage ripple.

The integration tests also found problems with charge-pump polarity, divider supply connections and the reference source period. These problems showed that clear interfaces and shared checking matter as much as local circuit design. A useful review checks the supply, ground, pin names and expected signal behaviour before a long simulation is started.

For future teamwork, each task should have a clear owner and a review partner. Design changes should be recorded with their reason and result. A shared final copy should contain the approved schematics, layout and test settings. This helps prevent one person from using an older version while another person is testing a newer one.

5.3 Diversity Statement of Team

This project needs several kinds of skill: analog design, digital logic, feedback analysis, simulation, layout and communication. A team can benefit when members bring different strengths and ways of solving problems. Careful checking, calculations and clear explanations deserve the same respect as the main circuit design.

A fair team gives every member a chance to explain ideas, ask questions and learn unfamiliar tasks. Technical decisions should be based on calculations and test results. Work should be divided fairly, and the final contribution record should reflect actual effort. Respect for different skills and fair participation are important throughout the project.

5.4 Log Book of Project Implementation

Table 16 records the weekly milestones, the role of each member and the team activities.

Date	Milestone achieved	Individual Role	Team Role	Comments
Week 7	Started the project; reviewed the PLL targets and prepared initial block schematics and device sizes.	All members: study the requirements and design the assigned blocks.	Agree on the work plan, divide the tasks and review block connections.	Project start; initial design ready.
Week 8	Built the PFD reset-delay cell and began charge-pump tests.	2106182, 2106150: PFD. 2106089, 2106150, 2106181: charge pump.	Review pulse width, dead zone and current mismatch.	Block testing began.
Week 9	Improved charge-pump matching, bias start-up and replica-loop stability.	2106089, 2106150, 2106181: charge pump and bias circuits.	Compare current and stability results; select the improved design.	Matching and stability improved.
Week 10	Verified VCO tuning, gain and phase noise; checked the divider.	2106150, 2106089: VCO. 2106181, 2106089: divider.	Confirm the tuning range and divide-by-16 operation.	VCO and divider checked.
Week 11	Rescaled the loop filter, connected the full PLL and checked lock, jitter, spur and power.	2106181, 2106182: filter. 2106150, 2106182: integration and PLL tests.	Review feedback polarity; fix integration problems and agree on final schematics.	System results reviewed.
Week 12	Completed block and top-level layouts; ran DRC and LVS.	All members: block layouts. 2106150, 2106182: top-level checks.	Review routing, pin names and layout connectivity.	Physical checks began.
Week 13	Corrected DRC/LVS errors and prepared the report and slides.	2106150, 2106182: final checks. All members: documentation.	Verify the final results and practise the presentation.	Design ready for demonstration.
Week 14	Demonstrated the PLL and presented the final results.	All members: explain their contributions.	Present the project, answer questions and finalise the report.	Final demonstration.

Table 16: Project implementation logbook from Week 7 to Week 14.

6 Communication to External Stakeholders (PO(j))

The main audience for this project is the course instructors and project evaluators. A technical reader needs the circuit structure, test conditions and evidence behind each result. A general reader needs a simple explanation of what the circuit does and why accurate timing matters. The report and demonstration should make both levels clear.

GitHub Link - <https://github.com/intmushfiq/Design-of-a-CMOS-Integer-16-Charge-Pump-PLL->

6.1 Executive Summary

Our team designed a clock circuit that turns a 75 MHz reference into a 1.20 GHz output. It can help electronic systems keep accurate timing. The design uses 45 nm CMOS technology and a 1.1 V supply. Simulation showed that it locked in about 0.61 microseconds and used 2.315 milliwatts of power. The layout passed the required design and connection checks. The project started in Week 7 and was demonstrated in Week 14. This is a simulation and layout project. A manufactured chip still needs testing and further careful verification before the design can be used in practical electronic products.

6.2 User Manual

This guide describes how to open and test the simulated PLL in Cadence. The required environment is Cadence Virtuoso with Spectre, the GPDK045 technology files, the project library and the saved testbenches. A compatible tool and PDK installation is needed to reproduce the results.

1. **Open the project.** Start Virtuoso in the project working directory. Check that the project library and GPDK045 library appear in Library Manager. Keep a backup before changing the final design.
2. **Check the library paths.** Check the project entries in cds.lib and the technology attachment. Open the top-level PLL schematic and confirm that the PFD, charge pump, loop filter, seven-stage VCO and divide-by-16 chain resolve correctly.
3. **Select the device models.** Open the saved ADE setup and confirm the GPDK045 Spectre model file and the appropriate typical process sections for the device types. Do not run with missing models or unresolved instances.
4. **Check the supply and reference.** Set the supply to 1.1 V and use the 75 MHz reference clock. Its period is about 13.333 ns. Check the source units, supply polarity, ground connections and control signal polarities.
5. **Check the feedback path.** Confirm that the divider output goes to the feedback input of the PFD. The division ratio must be 16. Confirm that the charge-pump output goes through the loop filter to the VCO control input.

6. **Run transient simulation.** Use the saved initial conditions and run a transient simulation long enough to include start-up and a locked interval; a 2 microsecond run covers the interval used in the report. Retain sufficiently small time steps and waveform resolution to capture the short PFD pulses.
7. **Check lock.** Plot the control voltage, VCO output, divider output and reference clock. The reported control voltage after lock is about 575.7 mV. The output should approach 1.20 GHz, and the divider output should approach 75 MHz. The reported lock time is 612.5 ns.
8. **Measure the locked behaviour.** Use a time window after lock; the report uses 0.8 to 2 microseconds for period jitter. Calculate output frequency, RMS period jitter, peak-to-peak control ripple and average supply power. The reported values are 1.214 ps jitter, 10.13 mV ripple and 2.315 mW power.
9. **Repeat the block checks.** Sweep the VCO control voltage from 0.2 to 0.9 V. Check charge-pump UP/DN matching over the same range. Use the saved oscillator noise setup for phase noise and the locked-output spectrum setup for reference spur. Compare against the report's -91.9 dBc/Hz at 1 MHz offset and -69.4 dBc reference spur.
10. **Check the two feedback loops.** Use the separate linear phase model to check the PLL bandwidth and phase margin. Use the charge-pump replica-loop stability setup to check that loop across the control range. The PLL phase margin and the replica-loop phase margin are different quantities.
11. **Verify the layout.** Open the final top-level layout. Run DRC with the correct rule deck, then extract connectivity and run LVS against the matching schematic. Read the error reports and correct any remaining violations before using the layout for later work.
12. **Save the evidence.** Save the final schematics, layout, ADE states, waveforms and verification reports. Label the results with the circuit version and test conditions so another user can reproduce them.

If the PLL does not lock, first check the supply, reference period, divider connections and charge-pump control polarity. If a simulation fails, read the error log and check model paths and initial conditions. If DRC or LVS fails, correct the reported geometry or connection errors. Change one cause at a time and repeat the affected test.

7 Project Management and Cost Analysis (PO(k))

No bill of materials or per-unit cost is given, because the project was carried out entirely in simulation and layout software. The Cadence tools and the GPDK045 design kit were provided by the department, and no hardware was purchased or fabricated, so no project cost was incurred. This section therefore covers the project schedule only.

7.1 Timeline of Project Implementation

The Gantt chart in Table 17 shows the eight-week schedule. Some activities overlap so that block revisions, layout and documentation can progress together.

X and light blue shading indicate scheduled work. D and dark blue shading mark the Week 14 demonstration.

Activity	W7	W8	W9	W10	W11	W12	W13	W14
Project planning and background study	X							
Block schematics and initial sizing	X							
PFD design and dead-zone tests	X	X						
Charge-pump matching and stability		X	X					
VCO tuning and noise verification			X	X				
Divider design and verification			X	X				
Loop-filter design and rescaling				X	X			
PLL integration and performance tests					X			
Block and top-level layout					X	X		
DRC and LVS checks and corrections						X	X	
Report and presentation preparation						X	X	X
Final demonstration								D

Table 17: Gantt chart from Week 7 to Week 14.

8 Future Work (PO(I))

The next priority is to check whether the good schematic results remain valid after layout and under wider operating conditions. The current work establishes a working nominal design, but further verification is needed before fabrication or practical use.

Post-layout verification. Extract the layout resistance and capacitance, then repeat lock-time, frequency, jitter, ripple, spur and power tests. Extra loading and coupling can change oscillator frequency and loop behaviour. Any changed result should be compared with the original specification.

Variation and reliability. Test the complete PLL at the relevant process corners and across the intended supply and temperature range. Run statistical mismatch tests on the charge pump, current reference and other sensitive blocks. This will show how much the results change between devices and operating conditions.

Tuning margin. Add tuning margin at both ends of the range so that 0.9–1.5 GHz is still covered under process, supply and temperature variation. Continue using the measured VCO gain when calculating the loop filter.

More realistic testing. Include a realistic reference clock, supply noise and output load. Check repeated power-up, different initial control voltages and unusual feedback conditions. These tests will help identify failure cases that an ideal reference and nominal start-up test can miss.

Circuit improvement. Investigate ways to reduce the control-voltage ripple and power while keeping the PFD free of a dead zone. Improve UP/DN pulse balance and current matching where possible. Any smaller devices or lower bias currents must be checked again for phase noise, stability and tuning range.

Hardware validation. If fabrication is pursued, move the design to a suitable manufacturing process and add pads, electrostatic protection, packaging and a test plan. Obtain real cost quotations. Then compare measured chip performance with the simulations rather than assuming they will agree. Further work also requires continued learning. Studying oscillator noise, statistical analog design, extraction and reliable simulation methods will help us address these next steps. Clear design notes and reproducible test setups will make that learning useful to future project teams.

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